Kernel Methods for the Computation of Quasi-Potentials in Stochastic Dynamical Systems

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Abstract

We present a data-driven, kernel-based framework for reconstructing the quasi-potential kernel framework for computing quasi-potentials of stochastic differential equations (SDEs) in the weak-noise regime. The method approximates a family of Koopman eigenfunctions in a reproducing-kernel Hilbert space and combines them into a closed-form surrogate whose gradient automatically satisfies a quadratic Hamilton–Jacobi ansatz. A composite loss simultaneously (i) minimises the Hamilton–Jacobi residual on a collocation grid, (ii) matches the Hessian of the surrogate to the solution of the Riccati equation at the equilibrium, and (iii) promotes sparsity through an ℓ_1 penalty on the kernel-mixture weights Direct scalar RKHS approximations and structured Koopman-coordinate surrogates combine Hamilton–Jacobi residual fitting with diffusion-dependent Riccati curvature at a stable equilibrium. Kernel-amplitude regularization and finite-dictionary selection provide complementary ways to control representation complexity. A conditional convergence theorem states the compactness, residual-control, and boundary comparison assumptions needed to identify the viscosity limit.

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Numerical experiments show that the analytic surrogate is inexpensive to evaluate, paving the way for efficient rare-event analysis and optimal-control applications in higher-dimensional, non-gradient stochastic systems recover two quartic reference potentials with polynomial kernels and reproduce a three-dimensional multiplicative-noise quadratic with a stable anchored kernel expansion. The three-dimensional fit achieves an RMSE of 1.35×10^{-9} on an 8000-point grid at $N = 200$. Independent tests, multiple training seeds for the two-dimensional systems, and a structured eight-dimensional experiment assess accuracy and computational cost. A comparison with restricted quadratic Koopman representations identifies approximation bias associated with transformed diffusion. All numerical results are accompanied by code, fitted coefficients, and native vector figures.

1 Introduction

Noise-induced transitions and rare escape phenomena are ubiquitous in physical, chemical, biological, and engineering systems [34, 59]. The Freidlin-Wentzell large deviation theory serves as a fundamental framework for analyzing rare events driven by weak noise [11, 21, 36, 48, 49]. Central to this theory is the concept of the action functional, which quantifies the likelihood of trajectories deviating from deterministic paths. The quasi-potential, defined as the infimum of the action functional, lies at the heart of large deviation theory. It characterizes the probability of a stochastic system fluctuating into a neighborhood of a specific state and governs key quantities such as mean first exit times and stationary distributions.

Despite its importance, computing the quasi-potential remains challenging. Traditional approaches can be categorized into three classes. Action Plot Methods [4]: These methods parameterize trajectories by integrating along unstable manifolds in the extended Hamiltonian phase space, starting from a small circle near a fixed point until exiting the basin of attraction. While effective for obtaining quasi-potential values along parameterized paths, they fail to establish a direct relationship between the quasi-potential and spatial coordinates. Geometric Minimum Action Method (GMAM) [33]: As a representative of string methods [13, 24, 55, 58], GMAM iteratively identifies the most probable path between a fixed point and a target state. However, it computes the quasi-potential only at discrete endpoints, lacking global structure. Ordered Upwind Method (OUM) [12, 20]: By discretizing the phase space and propagating quasi-potential values outward from fixed points, OUM achieves global quasi-potential reconstruction. Yet, its computational efficiency remains limited, especially in high-dimensional systems.

Recent advances in computational power and artificial intelligence have spurred the application of machine learning to stochastic dynamics. Techniques such as sparse regression [6, 42], Koopman operators [39, 56], and neural networks [15] have been leveraged for data-driven modeling of stochastic systems. In addition, some machine learning methods are also proposed to solve the problem of noise-induced transitions such as computing the most probable paths [43, 54] and the mean first exit times [45]. Notably, Lin et al. [46] proposed neural network-based frameworks to compute quasi-potentials for stochastic dynamical systems with additive Gaussian noise, while Li et al. [44] developed physics-informed neural networks (PINNs) to compute quasi-potentials by solving Hamilton-Jacobi equations.

Among machine learning paradigms, kernel methods stand out due to their strong nonlinear representation capabilities and computational efficiency. Notably, reproducing kernel Hilbert spaces (RKHS) [19] have established solid mathematical foundations for the analysis of dynamical systems [1, 5, 7–10, 22, 25–29, 31, 32, 35, 37–39, 41, 47] and

surrogate modeling (cf. [50] for a survey) as well as analyzing neural networks [52]. The idea of combining multiple kernels by learning a weighted mixture is known as *multiple kernel learning* (MKL) [2, 40] and has been studied extensively in the machine learning literature. Our approach is related to MKL in that we learn a sparse convex combination of base kernels, but it differs in that the objective couples kernel selection with PDE residual minimization and curvature matching, rather than regression or classification loss.

Position relative to existing approaches. Kernel collocation and Gaussian-process methods provide smooth representations, derivative evaluation, and nonlinear PDE residual fitting [16]. Physics-informed quasi-potential methods impose the Hamilton–Jacobi equation [44, 46], while kernel Koopman methods approximate dynamical eigenfunctions and spectra [57]. The present contribution couples a diffusion-dependent Riccati curvature target with direct scalar and Koopman-coordinate kernel surrogates, and specifies their distinct approximation properties. The Riccati term supplies local nontrivial geometry; global viscosity selection requires further conditions. Koopman coordinates give a structured nonnegative parametrization and linear inner transport fits, but their benefit over direct scalar RKHS fitting is problem dependent. Section 3 explains this distinction, and Appendix A provides a controlled comparison. Section 4 studies two- and three-dimensional systems. Neither the meshfree representation nor those examples establishes general high-dimensional scalability.

In this work, we present a kernel-based algorithm for quasi-potential computation. The structure of this paper is as follows: Section 2 introduces the definition and properties of the quasi-potential. Section 3 details the proposed kernel-driven methodology. Numerical examples in Section 4 validate the algorithm's accuracy and efficiency examine reconstruction accuracy and computational limitations, followed by concluding remarks.

2 Quasi-Potential and the Wentzel–Kramers–Brillouin Wentzel–Kramers–Brillouin Approximation in Stochastic Dynamical Systems

In this section, we examine dynamical systems governed by state variables $x_t \in \mathbb{R}^n$, evolving under the influence of Gaussian white noise with low intensity. The dynamics are described by the stochastic differential equation:

$$dx_t = b(x_t)\,dt + \sqrt{\varepsilon}\,\sigma(x_t)\,dB_t, \tag{1}$$

where $B_t \in \mathbb{R}^n$ denotes an n-dimensional standard Brownian motion, and ε is a small positive parameter quantifying the noise strength. The function $b(x) \in \mathbb{R}^n$ represents the deterministic drift term, which we assume has continuously differentiable derivatives. The diffusion coefficient is given by the matrix $a(x) = \sigma(x)\sigma(x)^\top$, where we assume $a(x)$ is uniformly positive definite on Ω (ensuring non-degeneracy of the noise).

We suppose $\bar{x}$ is an **asymptotically stable and hyperbolic** fixed point of the associated deterministic system, that is, $b(\bar{x}) = 0$ and all eigenvalues of its linearization matrix $J = \nabla b(\bar{x})$ have **strictly negative real parts**, i.e., $\mathrm{Re}(\lambda_i) < 0$ for all $i = 1, \ldots, n$. This ensures asymptotic stability in the sense that trajectories starting near $\bar{x}$ converge to $\bar{x}$ exponentially in time. Under stochastic perturbations, the process x_t explores the state space and converges to a stationary probability distribution additional global confinement and recurrence assumptions, the perturbed process admits a stationary density $p_s(x)$; local stability alone does not imply this global property. In the regime of weak noise ($\varepsilon \ll 1$),

this stationary distribution admits a Wentzel–Kramers–Brillouin (WKB)-type asymptotic approximation (see [17, 44, 51]):

$$p_s(x) \sim C(x)\, \exp\left\{-\varepsilon^{-1} V(x)\right\}, \tag{2}$$

where $C(x)$ is a slowly varying prefactor and $V(x)$ is the quasi-potential function, characterizing the exponential decay rate of the probability density. This form is termed a *WKB-type approximation* named after the WKB method in quantum mechanics, where solutions in the semiclassical limit $\hbar \to 0$ take the form $\psi(x) \sim A(x) \exp\{iS(x)/\hbar\}$. In both contexts, the exponential term dominates and captures the leading-order behavior, while the prefactor varies more slowly. In stochastic systems, the small parameter ε appears in the exponent, leading to sharp concentration of the probability distribution around stable points as $\varepsilon \to 0$. The quasi-potential $V(x)$ acts as an effective potential or energy landscape, quantifying the cost of rare fluctuations. The symbol "$\sim$" denotes logarithmic equivalence, meaning that

$$\lim_{\varepsilon \to 0} \frac{\log p_s(x)}{-\varepsilon^{-1} V(x)} = 1,$$

$$\lim_{\varepsilon \to 0} -\varepsilon \log p_s(x) = V(x),$$

which highlights that $V(x)$ governs the leading-order behavior in the small noise limit. after compatible normalization and in the relevant single-attractor asymptotic setting. In a multistable problem, the stationary rate function and the action value relative to one attractor must be distinguished.

The concept of the quasi-potential originates from large deviation theory [49]. It is defined via a variational principle involving the Freidlin–Wentzell Freidlin–Wentzell action functional $S_T[\phi]$, formulated over continuous trajectories $\phi(t)$ that connect a stable fixed point $\bar{x}$ to a target point x. Specifically, the quasi-potential is given by the nested infimum:

$$V(x) \equiv \inf_{T>0} \inf_{\phi \in C([0,T])} \left\{ S_T[\phi] \,:\, \phi(0) = \bar{x},\ \phi(T) = x \right\},$$

$$V(x) \equiv \inf_{T>0} \inf_{\phi \in AC([0,T])} \left\{ S_T[\phi] \,:\, \phi(0) = \bar{x},\ \phi(T) = x \right\}, \tag{3}$$

where $C([0,T])$ denotes the space of continuous functions on the interval $[0,T]$ $AC([0,T])$ denotes absolutely continuous paths; the action is infinite outside its admissible path class. The action functional $S_T[\phi]$, fundamental in the large deviation framework, is given by

$$S_T[\phi] = \int_0^T L(\phi(t), \dot{\phi}(t))\, dt, \tag{4}$$

with the Lagrangian $L(\phi, \dot{\phi})$ for the stochastic differential equation (1) defined as follows

$$L(\phi, \dot{\phi}) = \frac{1}{2} \left[\dot{\phi} - b(\phi)\right]^T a^{-1}(\phi) \left[\dot{\phi} - b(\phi)\right]. \tag{5}$$

This action functional quantifies the cost of deviations from the deterministic path and plays a central role in describing the asymptotic probability of rare events. In particular, for a fixed trajectory $\phi(t)$, the probability that the stochastic process x_t remains within a small tube around $\phi(t)$ over the time interval $[0,T]$ satisfies the large deviation estimate:

$$\mathbb{P}_{x_0} \left\{ \sup_{t \in [0,T]} |x_t - \phi(t)| < \delta \right\} \sim \exp\left\{-\varepsilon^{-1} S_T[\phi]\right\}, \tag{6}$$

for small $\delta > 0$, illustrating that the action functional governs the exponential rate of decay of such pathwise probabilities.

Note that the stationary probability distribution satisfies the time-independent Fokker–Planck equation

$$\mathcal{L}p_s(x) = -\sum_{i=1}^{n} \frac{\partial}{\partial x_i} b_i(x) p_s(x) + \sum_{i,j=1}^{n} \frac{\partial^2}{\partial x_i \partial x_j} a_{ij}(x) p_s(x) = 0,$$

Fokker–Planck equation

$$\mathcal{L}p_s(x) = -\sum_{i=1}^{n} \frac{\partial}{\partial x_i} \big(b_i(x) p_s(x)\big) + \frac{\varepsilon}{2} \sum_{i,j=1}^{n} \frac{\partial^2}{\partial x_i \partial x_j} \big(a_{ij}(x) p_s(x)\big) = 0,$$

where $\mathcal{L}$ is the Fokker–Planck Fokker–Planck operator. Substituting the WKB ansatz (2) into this equation and collecting the leading-order terms in ε yields the Hamilton–Jacobi Hamilton–Jacobi equation:

$$H(x, p) \equiv b^T p + \frac{1}{2} p^T a p = 0, \quad p \equiv \nabla V(x), \tag{7}$$

which is a first-order nonlinear partial differential equation for the quasi-potential V. It is observed that this Hamiltonian is the Legendre transform of the Lagrangian (5). According to the definition (3), the quasi-potential at the fixed point satisfies $V(\bar{x}) = 0$. Since the quasi-potential function attains minimal value at the stable fixed point, the gradient of the quasi-potential is also zero at the stable point, i.e., $\nabla V(\bar{x}) = 0$.

If $b(x) = -\nabla U(x)$ (i.e., a gradient system) and $a(x) = I$ (i.e., additive noise), then the within the relevant attraction basin the quasi-potential function equals twice the potential: $V(x) = 2U(x)$ is $V(x) = 2[U(x) - U(\bar{x})]$, under the usual connecting-path assumptions. In this special case, the quasi-potential admits an analytical expression.

Assume that the second-order approximation of the quasi-potential function at the fixed point $\bar{x}$ is given by

$$\begin{aligned} V(x) &\approx V(\bar{x}) + \nabla V(\bar{x})(x - \bar{x}) + \frac{1}{2}(x - \bar{x})^T \nabla^2 V(\bar{x})(x - \bar{x}) \\ &= \frac{1}{2}(x - \bar{x})^T Z (x - \bar{x}), \end{aligned}$$

where $Z = \nabla^2 V(\bar{x})$. Consequently, its gradient is approximated as

$$\nabla V(x) \approx Z(x - \bar{x}).$$

Substituting the gradient approximation into the Hamilton–Jacobi Hamilton–Jacobi equation (7) yields

$$b(x)^T Z(x - \bar{x}) + \frac{1}{2}(x - \bar{x})^T Z^T a(x) Z(x - \bar{x}) \approx 0. \tag{8}$$

Near the fixed point, we also approximate the drift $b(x)$ and the diffusion $a(x)$ as

$$b(x) \approx F(x - \bar{x}), \ a(x) \approx A,$$

where $F \equiv \nabla b(\bar{x}), A \equiv a(\bar{x})$. Substituting the linearized $b(x)$ and $a(x)$ into Eq. (8), we have

$$(x - \bar{x})^T (F^T Z + \frac{1}{2} Z^T A Z)(x - \bar{x}) \approx 0. \tag{9}$$

Note that it is equivalent to the following equation.

$$(x-\bar{x})^T(F^TZ+ZF+Z^TAZ)(x-\bar{x})\approx 0. \tag{10}$$

Since the matrix $F^TZ+ZF+Z^TAZ$ is a symmetric matrix, it must be diagonalizable. Due to the arbitrariness of $x-\bar{x}$, the eigenvalues of the matrix must all be 0. That is, each element of the matrix is 0. Therefore, the Hessian matrix $Z=\nabla^2V(\bar{x})$ satisfies the algebraic Riccati equation:

$$F^TZ+ZF+Z^TAZ=0 \tag{11}$$

or the equivalent form:

$$FZ^{-1}+Z^{-1}F^T+A=0. \tag{12}$$

The principal challenge in computing the quasi-potential from the PDE (7) lies in avoiding trivial solutions. To address this, we employ kernel methods to solve the PDE (7), and use equation (12) to impose that the Hessian of V at the origin coincides with the matrix Z solving (12).

It is worth emphasizing that the quasi-potential plays a pivotal role not only in large deviation theory but also in stochastic dynamics, as it is linked to various physical quantities. For example, the expected time to exit the domain of attraction of a fixed point—known point—known as the mean exit time—is time—is exponentially determined by the minimum of the quasi-potential on the boundary of the domain. The exit location distribution and the invariant measure can also be approximated based on the quasi-potential. Furthermore, once the quasi-potential function is known, the most probable transition path connecting two steady states can be derived from the quasi-potential. Typically, the most probable path passes through a saddle point, with the latter half of the trajectory beyond the saddle point following its unstable manifold toward the destination steady state. The first half of the path is obtained by solving the equation

$$\dot{x}=b(x)+a(x)\nabla V(x). \tag{13}$$

The boundary conditions are $x(-\infty)=\bar{x}$ and $x(\infty)=x_{sa}$, where x_{sa} denotes the saddle point. In practice, Equation (13) is obtained by substituting $p=\nabla V(x)$ into the state equation $\dot{x}=b(x)+a(x)p$ of the Hamiltonian system corresponding to the Hamilton-Jacobi Equation (7). Therefore, the most probable path is the projection of a heteroclinic orbit in the four-dimensional $2n$-dimensional Hamiltonian phase space onto the two-dimensional n-dimensional state space. For further properties of the quasi-potential, see [49].

Remark 2.1. When $\varepsilon=0$ (i.e. when there is no noise), our work will reduce to constructing the Lyapunov function for the unforced system $\dot{x}=b(x)$ [23, 41]

3 Method for Approximating the Quasi-Potential

To approximate the quasi-potential, we exploit the spectral structure of We consider both direct scalar kernel approximations and a structured spectral representation. The latter exploits the Koopman operator; its construction is developed first, and the direct finite-feature implementation is given in Section 3.1. Specifically, we construct eigenfunctions $\varphi_i(x)$ satisfying the linear advection-type partial differential equations:

$$\nabla\varphi_i(x)\cdot b(x)=\lambda_i\varphi_i(x),\quad \varphi_i(\bar{x})=0,\quad \nabla\varphi_i(\bar{x})=e_i,$$

$$\nabla\varphi_i(x)\cdot b(x)=\lambda_i\varphi_i(x),\quad \varphi_i(\bar{x})=0,\quad \nabla\varphi_i(\bar{x})=\ell_i, \tag{14}$$

where λ_i are eigenvalues of the linearization $J = \nabla b(\bar{x})$ at the fixed point $\bar{x}$, and e_i are the standard basis vectors . ℓ_i are left eigenvectors, $J^\top \ell_i = \lambda_i \ell_i$. Standard basis vectors give this normalization only in suitable coordinates. Complex conjugate modes are handled through their real and imaginary parts; smooth independent eigenfunctions on an entire basin require additional dynamical assumptions. These eigenfunctions correspond to the Koopman eigenfunctions associated with the equilibrium dynamics and encode local geometric and dynamical information such as the structure of stable and unstable manifolds.

Let $\Phi(x) = (\varphi_1(x) \cdots \varphi_n(x))^T$, and define the matrix W whose columns are the eigenvectors w_i of J, satisfying $Jw_i = \lambda_i w_i$. We then define a quadratic approximation to the quasi-potential:

$$V_{\text{mid}}(x) = \tfrac{1}{2}\Phi(x)^T P_{\text{mid}} \Phi(x),$$

$\Phi = (\varphi_1, \ldots, \varphi_n)^\top$ in a real coordinate representation, and let S be the invertible matrix of prescribed gradients at $\bar{x}$ (rows $\ell_i^\top$, or corresponding real blocks). We define

$$V_{\text{mid}}(x) = \tfrac{1}{2}\Phi(x)^\top P_{\text{mid}} \Phi(x), \qquad P_{\text{mid}} = S^{-\top} P_{\text{loc}} S^{-1}, \tag{15}$$

with the symmetric matrix [1]:

$$P_{\text{mid}} = W \,\text{diag}(-2\lambda_i)\, W^{-1}.$$

where P_{loc} is the positive-definite physical Riccati Hessian. This matrix is symmetric and includes the diffusion information. If the value and gradient normalizations of Φ hold exactly, the Hessian of V_{mid} is exactly P_{loc}; the curvature penalty remains useful for approximate enforcement. A similarity transform of eigenvalues alone is not a general formula for this Hessian.

To represent $\varphi_i(x)$, we employ a data-driven kernel-based ansatz:

$$\varphi_i(x) = \sum_{p=1}^{N} \alpha_p^{(i)}\, k_\beta(x, x_p) + (x - \bar{x}) \cdot w_i,$$

$$\varphi_i(x) = \sum_{p=1}^{N} \alpha_p^{(i)}\, k_\beta(x, x_p) + (x - \bar{x}) \cdot \ell_i, \tag{16}$$

where the point set $\{x_p\}_{p=1}^N$ denotes the collocation set which is used for training, and $k_\beta(x,t)$ is $k_\beta = \sum_j \beta_j^2 k_j$, a convex combination of base kernels (e.g., polynomials and radial basis functions), parameterized by a non-negative weight vector β with nonnegative amplitudes β satisfying $\sum_j \beta_j^2 = 1$. This kernel formulation ensures flexibility and regularization, and it circumvents the need to construct the Koopman operator explicitly.

The coefficients α_p are determined by minimizing the regularized empirical loss functional using the representer theorem a regularized finite-section collocation fit. A derivative-loss representer theorem involves the corresponding derivative functionals and does not by itself identify the quadratic eigenfunction surrogate with a scalar RKHS minimizer. To determine the coefficients $\alpha^{(i)} = (\alpha_1^{(i)}, \ldots, \alpha_N^{(i)})$, we enforce the Koopman eigenfunction equation at the collocation points. Specifically, for each $i = 1, \ldots, n$, we impose the conditions

$$\varphi_i(\bar{x}) = 0, \qquad \nabla\varphi_i(\bar{x}) = e_i,$$

[1] Alternatively, one can set $P_{\text{mid}} = \frac{1}{2}(P_u + P_\ell)$, where P_u and P_ℓ solve Riccati-type equations: $P\Lambda + \Lambda^\top P + PRP = 0$, for different choices of R.

$$\varphi_i(\bar{x}) = 0, \qquad \nabla\varphi_i(\bar{x}) = \ell_i,$$

together with the PDE constraint

$$\nabla\varphi_i(x_p) \cdot b(x_p) = \lambda_i\, \varphi_i(x_p), \qquad p = 1, \dots, N.$$

Substituting the RKHS ansatz (16) into these conditions yields a linear system for the coefficients $\alpha^{(i)}$, which we solve in a regularized least–squares sense:

$$\min_{\alpha^{(i)}} \sum_{p=1}^{N} \left(\nabla\varphi_i(x_p) \cdot b(x_p) - \lambda_i\, \varphi_i(x_p)\right)^2 \; + \; \eta\, \|\varphi_i\|^2_{\mathcal{H}_k}, \qquad \|\varphi_i\|^2_{\mathcal{H}_k} = \alpha^{(i)T} G_\beta \alpha^{(i)}. \tag{17}$$

Here $(G_\beta)_{pq} = k_\beta(x_p, x_q)$ is the kernel Gram matrix. The solution of this regularized linear system provides the coefficients $\alpha^{(i)}$ used to construct $\Phi(x)$ and the quasi-potential surrogate $V_{\mathrm{mid}}(x)$.

Remark 3.1 (RKHS norm and the linear correction term)**.** The ansatz (16) decomposes φ_i into an RKHS part $\sum_p \alpha_p^{(i)} k_\beta(x, x_p)$ and a fixed linear term $(x - \bar{x}) \cdot w_i$ $(x - \bar{x}) \cdot \ell_i$. The RKHS norm $\|\varphi_i\|^2_{\mathcal{H}_k} = \alpha^{(i)T} G_\beta \alpha^{(i)}$ penalizes only the learnable kernel component, not the linear correction. This is intentional: the linear term is analytically prescribed by the boundary condition $\nabla\varphi_i(\bar{x}) = e_i$ $\nabla\varphi_i(\bar{x}) = \ell_i$ and carries zero free parameters, so it should not be regularized. Penalizing it would bias the solution away from the exact local linearization.

To determine optimal kernel weights β, we minimize the following regularized loss function, where $P_{\mathrm{loc}} \in \mathbb{R}^{n\times n}$ denotes the local curvature target matrix—specifically, the positive-definite solution of the algebraic Riccati equation (12) evaluated at $\bar{x}$:

$$L(\beta) = \underbrace{\sum_p \left[b^T(x_p)\nabla V_{\mathrm{mid}}(x_p) + \frac{1}{2}\nabla V_{\mathrm{mid}}(x_p)^T a(x_p) \nabla V_{\mathrm{mid}}(x_p) \right]^2}_{L_{\mathrm{HJ}}} + \gamma \underbrace{\|\nabla^2 V_{\mathrm{mid}}(\bar{x}) - P_{\mathrm{loc}}\|_F^2}_{L_{\mathrm{curv}}}$$

$$+ \lambda_1 \sum_j |\beta_j| + \eta \sum_i \|\varphi_i\|^2_{\mathcal{H}_k}, \tag{18}$$

subject to the constraints $\beta_j \geq 0$ and $\sum_j \beta_j = 1$ $\sum_j \beta_j^2 = 1$. Here $\|\cdot\|_F$ denotes the Frobenius norm. The first term L_{HJ} enforces consistency with the Hamilton–Jacobi structure; the second term L_{curv} aligns the local curvature of the approximation with that of the Riccati solution at $\bar{x}$. The remaining terms promote sparsity and RKHS regularity.

Kernel amplitudes versus mixture weights. We use nonnegative amplitudes β_j with $\sum_j \beta_j^2 = 1$ and define the actual mixture weights as $w_j = \beta_j^2$. Thus $k_\beta = \sum_j \beta_j^2 k_j$ is a convex kernel combination, while the sparsity penalty is applied to the amplitudes. It is nonconstant:

$$1 \leq \sum_j \beta_j \leq \sqrt{L}.$$

Its minimum is attained by a single active amplitude, so it encourages concentrated mixtures; it does not guarantee that a nonconvex optimizer finds a particular sparse model. In contrast, an ℓ_1 penalty on simplex weights $w_j \geq 0$, $\sum_j w_j = 1$ is constant and has no sparsifying effect. The distinction is essential. Pruning small weights or amplitudes is an additional heuristic, followed by renormalization and refitting. In a single-component model the active amplitude and its weight both equal one. The discrete polynomial-degree selection used in Section 4 is specified separately from continuous mixture optimization.

Algorithmic structure. In practice, the computation proceeds in two stages. For a fixed choice of kernel weights β, the coefficients $\alpha^{(i)}$ are obtained by solving the regularised linear systems (17), yielding the Koopman eigenfunction approximations φ_i and hence $\Phi(x)$. In a second, outer optimisation loop, the kernel mixture coefficients β are updated by minimising the composite loss (18), while at each iteration the functions φ_i are recomputed from the updated kernel k_β. This bi-level structure (inner linear solves for $\alpha^{(i)}$, outer nonlinear optimisation for β) matches the implementation used in the numerical experiments of Section 4 separates the inner transport objective from the outer HJ objective. The inner and outer objectives play different roles when comparing the hybrid construction with a direct scalar fit.

Kernel dictionary. Throughout our numerical experiments, we employ a hybrid The general framework admits a dictionary combining Gaussian radial basis function (RBF) kernels and polynomial kernels. Section 4 examines polynomial components, and Appendix A evaluates Gaussian kernels. This choice is motivated by complementary approximation properties: RBF kernels excel at local interpolation and capturing rapid variations near critical points, while polynomial kernels naturally encode global algebraic structure characteristic of quasi-potentials. In our previous work [30], this RBF+polynomial combination dictionaries have demonstrated robust performance across diverse dynamical systems benchmarks . The dynamical-systems benchmarks [30]. Applied to the normalized amplitudes specified above, the ℓ_1 sparsity penalty term in the composite loss (18) acts as an automatic model selection mechanism analogous to encourages concentrated kernel mixtures, in the spirit of MKL [2, 40]: kernels that do not contribute meaningfully to the approximation have their weights driven to zero during optimization small components may be suppressed during fitting and removed by the stated pruning rule; exact sparse recovery is not guaranteed. In principle, the framework can accommodate much larger and more diverse kernel dictionaries; in [47] the same ℓ_1-based sparsification mechanism was used with an extensive kernel library to learn 132 chaotic dynamical systems. The methodology naturally extends to other kernel families (e.g., Matérn, rational quadratic, or domain-specific kernels) when more complex functional structures are anticipated.

Theoretical guarantee Role and limitations of the Koopman representation. The composite loss (18) couples three ingredients:

i. a Hamilton–Jacobi residual that enforces the global large-deviation structure,

ii. an ℓ_1 / RKHS regulariser that controls capacity, and

iii. a Riccati–curvature term that anchors the surrogate to the exact quadratic well at the attractor.

The next result shows that A direct comparator is $V_D(x) = \frac{1}{2}(x-\bar{x})^\top P_{\mathrm{loc}}(x-\bar{x}) + f_k(x)$, where the kernel correction can be constrained to have zero value, gradient, and Hessian at $\bar{x}$. Such a scalar model can minimize the same HJ residual without first learning eigenfunctions. Both methods permit analytic differentiation. The spectral model instead fits n transport equations, provides nonnegativity when $P_{\mathrm{mid}} \succ 0$, and supplies dynamical coordinates that may be reused in other tasks. These are structural advantages, not a proof of better accuracy or lower total cost.

In real coordinates with $D\Phi(\bar{x}) = I$, suppose $D\Phi\, b = F\Phi$ exactly and set $V_K = \frac{1}{2}\Phi^\top P \Phi$. Then

$$H(x, \nabla V_K) = \tfrac{1}{2}\Phi^\top \big(F^\top P + PF + P a_\Phi(x) P\big)\Phi, \quad a_\Phi = D\Phi\, a\, D\Phi^\top. \tag{19}$$

The transformed diffusion is generally state dependent, so local Riccati matching does not make this expression vanish everywhere. For example, $b(x) = -x(1 + x^2)$, under the standard assumptions of kernel-ridge learning and an appropriate schedule for the regularisation weights, these ingredients are sufficient to drive the surrogate V_N to the true $a(x) = 1 + x^2$ has $V(x) = x^2$ and the normalized eigenfunction $\varphi = x/\sqrt{1 + x^2}$ with eigenvalue -1. The quadratic spectral candidate is $V_K = \varphi^2 = x^2/(1 + x^2)$, with

$$H(x, V_K') = -\frac{2x^4(2 + x^2)}{(1 + x^2)^3}.$$

It matches the correct Hessian at zero but is not the quasi-potential V as the number of collocation points N grows. In particular, the Hessian of V_N at the equilibrium converges to the Riccati matrix P, giving a second-order match that accelerates the finite-sample error decay observed . Thus a finite quadratic eigenfunction representation can have approximation bias even with exact eigenfunctions. Appendix A compares a direct scalar correction and this restricted spectral construction using common data and curvature constraints; the tests favor the direct construction within these specified model classes. The Gaussian-kernel models in that comparison complement the polynomial-kernel experiments in Section 4. We state the result and the proof for completeness

Theoretical guarantee and solution selection. The composite loss (18) couples a Hamilton–Jacobi residual, a regularizer, and a Riccati-curvature term. These ingredients address PDE fit, capacity, and local geometry, respectively. A convergence statement must also identify the action-selected solution. Strict convexity of $H(x, \cdot)$ does not supply this selection: for $b(x) = -x$ and $a = 1$, both $u = 0$ and $u = x^2$ solve the HJ equation and satisfy $u(0) = 0$. Curvature matching excludes the constant candidate locally but does not by itself establish global uniqueness. We therefore state the comparison, compactness, and residual-control requirements explicitly, following viscosity stability [18].

Theorem 3.2 (Consistency with Riccati–curvature matching Conditional consistency with Riccati-curvature matching)**.** *Let $V : \Omega \subset \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ be the unique viscosity solution of the stationary Hamilton–Jacobi equation*

$$H\big(x, \nabla V(x)\big) = 0, \qquad V(\bar{x}) = 0,$$

$D \subset \mathbb{R}^d$ be a bounded domain with compact closure. Suppose b, a are continuous, a is uniformly positive definite, and compatible continuous boundary data g select the desired restriction V of the quasi-potential through

$$H(x, \nabla V) = 0 \quad \text{in } D, \qquad V = g \quad \text{on } \partial D.$$

and let $P \succ 0$ be the Riccati matrix

$$b'(\bar{x})^\top P + P\, b'(\bar{x}) \;=\; P\, a(\bar{x})\, P.$$

Assume Assume that this boundary-value problem has a continuous viscosity solution and a comparison principle in the relevant bounded continuous class. Let $V_N \in C^1(\overline{D})$ be a sequence of kernel-based surrogates satisfying:

1. *$b, a \in C^1(\Omega)$ and $a(x)$ is uniformly positive definite on Ω V_N are uniformly bounded and equicontinuous on $\overline{D}$;*

2. *k is a continuous, C^m $(m \geq 2)$* universal *kernel on Ω with RKHS $\mathcal{H}_k$ $\|V_N - g\|_{L^\infty(\partial D)} \to 0$;*

3. *Collocation points $\{x_p\}_{p=1}^N$ are drawn i.i.d. from a probability measure ρ whose support is $\overline{\Omega}$;*

4. *Regularisation parameters follow $\lambda_N \downarrow 0$, $N\lambda_N \uparrow \infty$, and a curvature weight γ_N satisfies $\gamma_N \downarrow 0$ with $\gamma_N/\lambda_N = O(1)$ for $r_N(x) = H(x, \nabla V_N(x))$, the collocation sets X_N have fill distance $h_N \to 0$, the residuals have a common modulus of continuity ω, and $\max_{x_p \in X_N} |r_N(x_p)| \to 0$.*

Then $V_N \to V$ uniformly on $\overline{D}$, and hence in $L^2(\rho)$ for every probability measure ρ on that closure.

For $N \in \mathbb{N}$, define the empirical objective:

$$\mathcal{L}_{N,\lambda,\gamma}(\Phi, \beta) := \frac{1}{N} \sum_{p=1}^{N} H\big(x_p, \nabla V_{(\Phi,\beta)}(x_p)\big)^2 + \gamma \big\| D^2 V_{(\Phi,\beta)}(\bar{x}) - P \big\|_F^2 + \lambda\big(\|\Phi\|_{\mathcal{H}_k}^2 + \|\beta\|_1\big),$$

and let $(\widehat{\Phi}_N, \widehat{\beta}_N)$ be any minimiser of (20). Set $V_N := V_{(\widehat{\Phi}_N, \widehat{\beta}_N)}$.

Then, as $N \to \infty$ If,

$$\|V_N - V\|_{L^2(\rho)} \xrightarrow{p} 0, \qquad \|\nabla V_N - \nabla V\|_{L^2(\rho)} \xrightarrow{p} 0, \qquad D^2 V_N(\bar{x}) \xrightarrow{p} P.$$

If$V \in C^m(\Omega)$, the convergence is uniform on compact subsets of Ω in addition, the functions are C^2 near $\bar{x}$ and satisfy $\|D^2 V_N(\bar{x}) - P\|_F \to 0$, their equilibrium curvature converges to P, where

$$b'(\bar{x})^\top P + P b'(\bar{x}) + P a(\bar{x}) P = 0, \qquad P \succ 0.$$

Gradient convergence requires additional compactness: if $V \in C^1$ and $\{\nabla V_N\}$ is relatively compact in $C(K)$ on every compact $K \Subset D$, then $\nabla V_N \to \nabla V$ uniformly on such K. None of these compactness or solution-selection conditions follows solely from small empirical loss.

In the theoretical analysis (Theorem 3.2), we use

Remark 3.3 (Notation Notation and empirical objective)**.** V_N to denote the N-sample approximationobtained by minimizing the empirical objective. In the numerical examples (Section 4), we use V_{mid} to denote the mid-level Koopman-based approximation constructed via Equation (15) denotes a sample-dependent approximation, V_{mid} the quadratic Koopman surrogate, and V_h or V_{quad} for specific baseline or hybrid approximations. These notational variants all refer to finite-sample approximations of the true quasi-potential V, distinguished only by their construction method or the context in which they appear V_{quad} particular numerical reconstructions. Write $\mathcal{R}(\Phi, \beta)$ for the nonnegative regularizer of the learnable kernel components and amplitudes. The normalized empirical objective is

$$\mathcal{L}_{N,\lambda,\gamma}(\Phi, \beta) = \frac{1}{N} \sum_{p=1}^{N} H(x_p, \nabla V_{(\Phi,\beta)}(x_p))^2 + \gamma \|D^2 V_{(\Phi,\beta)}(\bar{x}) - P\|_F^2 + \lambda \mathcal{R}(\Phi, \beta). \tag{20}$$

The theorem concerns sequences meeting its stated conditions, not arbitrary local or global minimizers of this nonconvex objective. Its boundary data are an additional solution-selection condition, not the single point constraint $V_N(\bar{x}) = 0$ used in a local fit.

Proof of Theorem 3.2. We follow the classical excess–risk decomposition of kernel ridge regression [14, 53], extended with curvature matching.

Step 1: Generalized representer theorem. The empirical loss in (20) depends on three types of evaluations: function values and gradients at the collocation points, and second derivatives at a single fixed point $\bar{x}$. By the **Step 1: Finite representations and their scope.** Kernel section expansions provide the numerical parametrizations. In an optimization over a scalar RKHS with bounded derivative evaluation functionals, a generalized representer theorem , this structure implies that V_N admits a finite-dimensional representation:

$$V_N(x) = \sum_{p=1}^{N}\sum_{j=1}^{n} c_{pj}\,\partial_{x_j}k(x,x_p) + \sum_{1\le j\le \ell\le n} d_{j\ell}\,\partial_{x_j x_\ell}k(x,\bar{x}),$$

with coefficients $\{c_{pj}\},\{d_{j\ell}\}$ determined by the optimiser. All derivatives belong to $\mathcal{H}_k$ by the C^m assumption on k, and the RKHS norm is bounded by

$$\|V_N\|_{\mathcal{H}_k} \;\le\; C\|\theta_N\|_2,$$

where θ_N stacks all the coefficients uses kernel derivatives associated with those functionals [16]. A quadratic form in RKHS eigenfunctions is not automatically the same scalar RKHS minimizer. The convergence argument below uses the regularity and compactness stated in the theorem, rather than making that identification. In particular, bounded second-derivative evaluation functionals require more kernel regularity than merely assuming that the kernel is C^2.

Step 2: Population risk and curvature penalty. Define the population risk:

$$R(V) \;:=\; \mathbb{E}_\rho\big[H\big(X,\nabla V(X)\big)^2\big], \qquad C(V) \;:=\; \big\|D^2V(\bar{x}) - P\big\|_F^2,$$

For any candidate u, define

$$R(u) = \mathbb{E}_\rho[H(X,\nabla u(X))^2], \qquad C(u) = \|D^2u(\bar{x}) - P\|_F^2.$$

and the regularised functional:

$$\mathcal{J}_{\lambda,\gamma}(V) := R(V) + \gamma C(V) + \lambda\|V\|_{\mathcal{H}_k}^2.$$

Since $a(x) \succ 0$, the Hamiltonian H is strictly convex in its second argument, ensuring that V is the unique viscosity solution and also the unique minimiser of $\mathcal{J}_{\lambda,\gamma}$ (as $C(V)=0$ and the regularisers vanish at V) The regularized population objective is $R(u)+\gamma C(u)+\lambda\mathcal{R}(u)$. Zero residual does not imply uniqueness, and a regularizer need not vanish at the true solution. Uniqueness of the viscosity limit will instead follow from the assumed boundary-value comparison principle.

Step 3: Empirical risk decomposition What empirical optimality gives. Define the empirical risk:

$$\widehat{R}_N(V) := \frac{1}{N}\sum_{p=1}^{N} H\big(x_p,\nabla V(x_p)\big)^2.$$

Let $\widehat{R}_N(u) = N^{-1}\sum_p H(x_p,\nabla u(x_p))^2$. If u_N minimizes the regularized empirical objective up to tolerance $\delta_N \ge 0$, comparison with any admissible w_N gives

$$\begin{aligned}\widehat{R}_N(u_N) + \gamma_N C(u_N) + \lambda_N\mathcal{R}(u_N) \le \widehat{R}_N(w_N) + \gamma_N C(w_N)\\ + \lambda_N\mathcal{R}(w_N) + \delta_N.\end{aligned} \tag{21}$$

Then V_N minimises the regularised empirical objective :

$$\widehat{\mathcal{J}}_N(V) := \widehat{R}_N(V) + \gamma_N C(V) + \lambda_N \|V\|^2_{\mathcal{H}_k}.$$

By standard excess risk decomposition:

$$R(V_N) - R(V) = \big[R(V_N) - \widehat{R}_N(V_N)\big] + \big[\widehat{R}_N(V_N) - \widehat{R}_N(V)\big] + \big[\widehat{R}_N(V) - R(V)\big].$$

The middle term is $\leq -\gamma_N C(V_N) - \lambda_N(\|V_N\|^2_{\mathcal{H}_k} - \|V\|^2_{\mathcal{H}_k})$ by optimality of V_N. Hence,

$$R(V_N) - R(V) + \gamma_N C(V_N) \;\leq\; 2 \sup_{V \in \mathcal{B}_N} \big|R(V) - \widehat{R}_N(V)\big|, \tag{22}$$

Define the comparison ball

$$\mathcal{B}_N := \Big\{ f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \leq \|V\|_{\mathcal{H}_k} + \lambda_N^{-1/2} \Big\},$$

where V denotes the true quasi-potential (as in the Theorem statement), so that $V_N \in \mathcal{B}_N$ with high probability All comparator regularization and optimization terms must be retained. This inequality explains the need for approximation and optimization control; it does not give a gradient-error bound. The comparison with kernel-ridge excess-risk analysis [14, 53] is methodological, not a transfer of its rates to the nonlinear HJ loss.

Step 4: Uniform deviation bound From samples to a uniform residual. Since k is bounded and smooth on compact Ω, and $H(x, \cdot)$ is Lipschitz in $\nabla V(x)$, the Lipschitz constant over $\mathcal{B}_N$ is $O(\lambda_N^{-1/2})$. Talagrand–Bousquet inequality gives:

$$\sup_{V \in \mathcal{B}_N} \big|R(V) - \widehat{R}_N(V)\big| \leq C \left(\sqrt{\frac{\log(1/\delta)}{N}}\, \lambda_N^{-1/2} + \frac{\log(1/\delta)}{N} \right)$$

For each $x \in \overline{D}$, choose a nearest collocation point. The common modulus yields

$$\|r_N\|_{L^\infty(D)} \leq \max_{x_p \in X_N} |r_N(x_p)| + \omega(h_N) \longrightarrow 0. \tag{23}$$

with high probability. Therefore,

$$\sup_{V \in \mathcal{B}_N} \big|R(V) - \widehat{R}_N(V)\big| = O_p\left(\sqrt{\frac{1}{N\lambda_N}} \right).$$

Here, the notation $\mathcal{O}_p(\epsilon_N)$ means that the random variable is bounded in probability at rate ϵ_N, i.e., for all $\delta > 0$ there exists $M > 0$ such that $\mathbb{P}(|Z_N| > M\epsilon_N) < \delta$ for all sufficiently large N This is a stronger requirement than $\widehat{R}_N(V_N) \to 0$. For example, $\max_p |r_N(x_p)| \leq \sqrt{N\widehat{R}_N(V_N)}$ gives the sufficient, but not necessary, condition $N\widehat{R}_N(V_N) \to 0$ when combined with uniform residual regularity. A population-risk route would need a uniform generalization estimate and sufficient control of residual spikes; no such rate is asserted here.

Step 5: Convergence in risk and curvature Compactness.

The assumption $N\lambda_N \to \infty$ implies that the RHS of (22) vanishes in probability. Since $\gamma_N/\lambda_N = O(1)$ and $\gamma_N \to 0$, it follows that:

$$R(V_N) - R(V) \to 0, \qquad C(V_N) \to 0 \quad \text{in probability.}$$

Uniform boundedness and equicontinuity give, by Arzelà–Ascoli, a uniformly convergent subsequence $V_{N_j} \to u$ on $\overline{D}$. The boundary assumption implies $u = g$ on ∂D. In particular, compactness is an independent hypothesis; kernel smoothness alone does not provide a uniform bound on fitted derivatives as regularization vanishes.

Step 6: Convergence in L^2 and gradient Viscosity identification.
The coercivity of $a(x)$ yields:

$$\tfrac{1}{2}\lambda_{\min}(a)\,\|\nabla V_N - \nabla V\|^2_{L^2(\rho)} \;\le\; R(V_N) - R(V),$$

If a C^1 function ϕ touches u from above at an interior point x, a small strictifying perturbation and uniform convergence produce local maxima of $V_{N_j} - \phi$ at $x_j \to x$. There,

$$\nabla V_{N_j}(x_j) = \nabla\phi(x_j), \qquad H(x_j, \nabla\phi(x_j)) = r_{N_j}(x_j) \to 0.$$

hence $\nabla V_N \to \nabla V$ Continuity of H gives the subsolution inequality. The same argument with local minima gives the supersolution inequality. Thus u is a viscosity solution with data g. Comparison identifies $u = V$. Every subsequential limit has this value, so the entire sequence converges uniformly and therefore in $L^2(\rho)$, and $V_N \to V$ in $L^2(\rho)$by Poincaré–Friedrichs on bounded Ω.

Step 7: Hessian consistency. $C(V_N) \to 0$ implies

$$D^2 V_N(\bar{x}) \to P \quad \text{in Frobenius norm, and hence in probability.}$$

The additional curvature hypothesis gives the stated Hessian convergence directly. For a penalized fit it is sufficient that the attained nonnegative objective in (20) is $o(\gamma_N)$, because

$$C(V_N) \le \mathcal{L}_{N,\lambda_N,\gamma_N}(V_N)/\gamma_N.$$

Merely taking $\gamma_N \to 0$ does not imply this conclusion. Exact curvature constraints are an alternative.

Step 8: Uniform convergence (optional) The additional gradient conclusion. If $V \in C^m(\Omega)$ and $k \in C^m$, the H_k norm controls m derivatives. By Sobolev embedding or Arzelà–Ascoli, convergence is uniform on compact subsets:

$$V_N \to V \quad \text{in } C^{m-1}(K) \quad \text{for all compact } K \Subset \Omega,$$

where $K \Subset \Omega$ implies that Under the stated gradient compactness assumption, any subsequence has a further subsequence whose gradients converge uniformly on K is a compact subsetwhose closure lies strictly inside Ω. This ensures that the convergence statements avoid potential boundary irregularities, which is standard in the analysisof convergence in function spaces.
to some q. Integration against compactly supported test functions, together with uniform convergence of V_N, identifies q as the distributional gradient of V, hence as ∇V when $V \in C^1$. Uniqueness of this limit gives convergence on every compact subset. No gradient convergence is asserted across a nonsmooth quasi-potential without further analysis. □

Theorem 3.2 establishes that the learned

Interpretation of the assumptions. For a quasi-potential V_N converges to the true solution V of the Hamilton–Jacobi equation in both value and gradient. In particular, the penalty term $\|D^2V_N(\bar{x}) - P\|_F^2$ enforces curvature consistency at the stable equilibrium $\bar{x}$. To clarify the necessity and interpretation of this Riccati-based curvature constraint, we include the following corollary , which shows that any sufficiently regular solution of the Hamilton–Jacobi equation must satisfy this Riccati condition locally at $\bar{x}$. , the boundary or admissibility problem must select the action value in (3). A truncated, punctured basin with compatible action data is one possible setting, provided its comparison principle is established. The local experimental fits do not establish those conditions and are therefore assessed through numerical diagnostics rather than claimed as instances of an unconditional consistency theorem. Likewise, the monotone-scheme theorem of Barles and Souganidis [3] cannot be applied directly: monotonicity of the present nonlinear kernel least-squares scheme has not been proved. A fixed quadratic Koopman class need not be dense in the required derivative topology even when the scalar kernel is universal. These distinctions preserve the role of Riccati matching without equating local curvature with global solution selection.

The following corollary records the local Riccati condition for a smooth solution with a nondegenerate minimum. It is separate from the global viscosity-selection question.

Corollary 3.4 (Local Consistency with the Hamilton–Jacobi Equation)**.** *Let $V \in C^2(\Omega)$ solve*

$$H(x, \nabla V(x)) = f(x)^\top \nabla V(x) + \tfrac{1}{2}\nabla V(x)^\top a(x)\, \nabla V(x) = 0, \qquad V(\bar{x}) = 0,$$

with $\bar{x}$ a globally asymptotically stable equilibrium of f. $f = b$, $b \in C^1$, $a(\bar{x}) \succ 0$, $Db(\bar{x})$ Hurwitz, $\nabla V(\bar{x}) = 0$, and $D^2V(\bar{x}) \succ 0$. The positive-Hessian condition excludes the constant HJ solution. Then the Hessian of V at $\bar{x}$ equals the Riccati matrix P:

$$D^2V(\bar{x}) = P,$$

where P is the unique positive-definite solution to the algebraic Riccati equation

$$b'(\bar{x})^\top P + P\, b'(\bar{x}) = P\, a(\bar{x})\, P.$$

$$b'(\bar{x})^\top P + P\, b'(\bar{x}) + P\, a(\bar{x})\, P = 0.$$

Proof. Near the equilibrium point $\bar{x}$, write the Taylor expansion

$$V(x) = \tfrac{1}{2}(x - \bar{x})^\top Z (x - \bar{x}) + o\big(\|x - \bar{x}\|^2\big), \qquad Z = D^2V(\bar{x}).$$

Linearise the drift and diffusion at $\bar{x}$:

$$f(x) = b'(\bar{x})(x - \bar{x}) + o(\|x - \bar{x}\|), \qquad a(x) = a(\bar{x}) + o(1).$$

Thus the gradient satisfies $\nabla V(x) = Z(x - \bar{x}) + o(\|x - \bar{x}\|)$. Insert these expansions into the Hamilton–Jacobi Hamilton–Jacobi equation:

$$H(x, \nabla V(x)) = f(x)^\top \nabla V(x) + \tfrac{1}{2}\nabla V(x)^\top a(x)\, \nabla V(x) = 0.$$

Collecting second-order terms in $x - \bar{x}$ yields

$$(x - \bar{x})^\top \big(b'(\bar{x})^\top Z + Z b'(\bar{x}) + Z a(\bar{x}) Z\big)(x - \bar{x}) + o\big(\|x - \bar{x}\|^2\big) = 0.$$

Since this holds for all x near $\bar{x}$, the symmetric coefficient must vanish:

$$b'(\bar{x})^\top Z + Z b'(\bar{x}) + Z a(\bar{x}) Z = 0.$$

By standard results on algebraic Riccati equations (positive-definite $a(\bar{x})$ and stability of $b'(\bar{x})$), there is a unique positive-definite solution P of this equation. Hence Since $Z \succ 0$, multiplying by Z^{-1} gives the uniquely solvable Lyapunov equation for Z^{-1}; hence $Z = P$ and therefore $D^2V(\bar{x}) = P$. □

4 Examples

Computational complexity, memory, and dimensional scope. Let N be the collocation count, M the number of centers ($M = N$ in a full section expansion), n the state dimension, L the dictionary size, and T the number of outer kernel evaluations. For common radial kernels, constructing values and first derivatives costs $O(LNMn)$; retaining all derivative arrays uses $O(LNMn)$ storage. A single dense $N \times N$ Gram matrix requires $8N^2$ bytes in float64. The often-quoted $O(N^2n)$ is an assembly cost, not the cost of solving a dense system: factorization requires $O(N^3)$ time and $O(N^2)$ memory. Initial computation of the Riccati target through a dense Lyapunov solve costs $O(n^3)$.

3.1 Example 1: The two-Dimensional Maier–Stein System

For n independent scalar eigenfunction fits with M centers, dense least-squares work can be $O(n(NM^2 + M^3))$ per kernel choice, in addition to assembly. Factorizations can sometimes be reused, but cannot generally be shared across distinct eigenvalues or updated kernels. Real block formulations for complex eigenpairs may require coupled solves. A generic fully coupled system has Nn rows and Mn columns: storing its design and normal matrices costs $O(NMn^2 + M^2n^2)$, and forming/factoring normal equations can cost $O(NM^2n^3 + M^3n^3)$. Exploiting spectral or block structure reduces this upper bound. The implementation in Appendix A uses the coupled real form.

We consider the Maier–Stein system whose deterministic part is

$$\dot{x} = F(x) = \begin{pmatrix} x_1 - x_1^3 - x_1x_2^2 \\ -(1 + x_1^2)x_2 \end{pmatrix},$$

which features a stable fixed point at $\bar{x} = (-1, 0)$. Throughout this example, we assume the diffusion matrix is the identity, i.e., $a(x) \equiv I$. Note that the corresponding deterministic system is a gradient systemwith A direct scalar M-coefficient HJ fit has nonlinear least-squares steps costing approximately $O(NM^2+M^3)$, plus gradient/residual assembly. Dense diffusion contributes $O(Nn^2)$ matrix-vector work, excluding coefficient evaluation. For structured radial derivatives, evaluating a trained spectral potential and gradient at one point costs approximately $O(LMn^2 + n^2)$, versus $O(LMn + n^2)$ for a scalar section expansion. Dense Taylor-Hessian contractions in the exact-curvature direct implementation of Appendix A add $O(Mn^2)$ work per point. The outer selection loop repeats inner work T times. Derivative workspaces, optimizer arrays, and validation evaluation must therefore be included in practical budgets.

Kernel sparsification reduces L but does not by itself reduce N or M. Nor does algebraic dependence on dimension in an individual operation remove the sample-coverage burden: a tensor grid with q points per coordinate has $N = q^n$. The examples in Section 4 cover dimensions two and three. Appendix A studies a structured eight-dimensional system and reports timing and storage measurements; these tests do not establish dimension-independent scalability. Sparse centers, structured linear algebra, and reduced-order models are possible future improvements, rather than established performance guarantees of the current method.

3.1 Direct scalar implementation and stable finite features

The HJ equation can be fitted directly in a scalar kernel space while retaining the same local Riccati information. This option separates potential approximation from the potential

$$U(x) = \frac{1}{4}(x_1^2 - 1)^2 + \frac{1}{2}(x_1^2 + 1)x_2^2.$$

The associated quasi-potential in this case is given by (here and throughout the examples, V_{true} denotes the analytically known quasi-potential, i.e., the true solution V from Theorem 3.2)

$$V_{\text{true}}(x) = 2U(x) = \frac{1}{2}(x_1^2 - 1)^2 + (x_1^2 + 1)x_2^2.$$

additional transport restriction imposed by a Koopman-coordinate model. Let $y = x - \bar{x}$. For polynomial kernels of degree r, use the finite-feature parametrization

$$V_r(x) = \tfrac{1}{2} y^\top P y + \sum_{3 \le |\nu| \le r} c_\nu y^\nu. \tag{24}$$

Linearization at the fixed point gives the Jacobian

$$J = DF(\bar{x}), \quad Jw_i = \lambda_i w_i,$$

where (λ_i, w_i) are the eigenpairs of J. We solve the Lyapunov equation

$$JZ + ZJ^\top = -I, \quad \Rightarrow \quad P_{\text{loc}} = Z^{-1},$$

yielding $P_{\text{loc}} = \nabla^2 V(\bar{x})$, Here ν is a multi-index and $y^\nu = \prod_i y_i^{\nu_i}$. The space of polynomials of degree at most r is the RKHS of the Hessian of the quasi-potential at the saddle inhomogeneous polynomial kernel $(1 + x \cdot z)^r$, with a degree-dependent inner product. Changing to centered monomials preserves this finite-dimensional function space. The coefficient penalty used in the implementation is explicitly Euclidean in these centered coordinates; it is not identified with the canonical polynomial-kernel norm.

We seek eigenfunctions $\varphi_i(x)$ satisfying the Koopman PDE:

$$\nabla\varphi_i(x) \cdot F(x) = \lambda_i \varphi_i(x), \quad \varphi_i(\bar{x}) = 0, \quad \nabla\varphi_i(\bar{x}) = e_i.$$

Restricting the correction to degrees at least three makes the value, gradient, and Hessian constraints exact. The coefficients minimize

$$\mathcal{J}(c) = \frac{1}{N} \sum_{p=1}^{N} H(x_p, \nabla V_r(x_p))^2 + \eta \|c\|_2^2. \tag{25}$$

Each φ_i is represented in an RKHS:

$$\varphi_i(x) = \sum_{p=1}^{N} \alpha_p^{(i)} k_\beta(x, x_p) + (x - \bar{x}) \cdot w_i, \quad k_\beta(x,t) = \sum_{j=1}^{L} \beta_j k_j(x,t).$$

Writing $g_p = \nabla V_r(x_p)$ and $D_{p\nu} = \nabla y_p^\nu$, the residual Jacobian is

$$\frac{\partial H(x_p, g_p)}{\partial c_\nu} = \big(b(x_p) + a(x_p) g_p\big)^\top D_{p\nu}.$$

On a collocation set $\{x_p\}_{p=1}^N$, we impose the constraints:

$$\begin{aligned}
&\varphi_i(\bar{x}) = 0, \\
&\nabla\varphi_i(\bar{x}) \cdot e_j = \delta_{ij}, \quad j = 1, 2, \\
&\nabla\varphi_i(x_p) \cdot F(x_p) = \lambda_i (x_p - \bar{x}) \cdot w_i, \quad p = 1, \dots, N.
\end{aligned}$$

Solving the regularized linear system yields the coefficients $\alpha^{(i)}$. This analytic derivative avoids differencing a nearly singular kernel linear solve. Columns are scaled by the training RMS of their derivatives for numerical conditioning; the transformation is undone when storing coefficients and evaluating the regularizer. Such scaling changes the optimization coordinates, not the fitted function or the objective.

Define $\Phi(x) = (\varphi_1(x), \varphi_2(x))^\top$, and construct the quasi-potential:

$$V_{\text{mid}}(x) = \tfrac{1}{2}\,\Phi(x)^T P_{\text{mid}} \Phi(x)$$

with

$$P_{\text{mid}} = W\,\text{diag}(-2\lambda_i)\,W^{-1}.$$

To determine β, we minimize the composite loss (18) (the same L_{HJ}, L_{curv} terms defined In two dimensions, the number of free coefficients in (24) is $\binom{r+2}{2} - 6$. The degree-two candidate has no correction and equals the Riccati baseline. Candidate degree selection uses independent HJ residuals, with the precise tolerance and continuation schedule specified in Section 3):

$$L(\beta) = L_{\text{HJ}} + \gamma L_{\text{curv}} + \lambda_1 \sum_{j=1}^{L} |\beta_j| + \eta \sum_{i=1}^{2} \|\varphi_i\|^2_{\mathcal{H}_k},$$

where

$$L_{\text{HJ}} = \sum_{p=1}^{N} \left[F(x_p) \cdot \nabla V_{\text{mid}}(x_p) + \tfrac{1}{2}\|\nabla V_{\text{mid}}(x_p)\|^2\right]^2,$$

$$L_{\text{curv}} = \|\nabla^2 V_{\text{mid}}(\bar{x}) - P_{\text{loc}}\|_F^2,$$

$$\|\varphi_i\|^2_{\mathcal{H}_k} = \alpha^{(i)\,T} G_\beta \alpha^{(i)}, \quad (G_\beta)_{pq} = k_\beta(x_p, x_q).$$

4. Analytic potential values are unnecessary for these choices.

Let β^* denote the optimal (possibly sparse) coefficient vector such that the true solution V can be well approximated within the model class spanned by the fixed dictionary. After obtaining β^*, we: Threshold small coefficients $\beta_j < 1/L$ to zero and renormalize. Recompute φ_i. Rescale V_{mid} via

$$s = \frac{\langle V_{\text{true}}, V_{\text{mid}}\rangle}{\|V_{\text{mid}}\|^2}, \quad \text{RMSE} = \sqrt{\frac{1}{n^2}\sum (sV_{\text{mid}} - V_{\text{true}})^2}.$$

An anchored quadratic kernel expansion. For the three-dimensional degree-two kernel, subtracting its value at the equilibrium gives a nine-dimensional function space. At $\bar{x} = 0$, write

$$q(x) = (x_1, x_2, x_3, x_1^2, x_2^2, x_3^2, x_1x_2, x_1x_3, x_2x_3)^\top, \qquad V_h(x) = q(x)^\top c.$$

By sweeping parameters λ_1, γ, and η, we observe trade-offs between sparsity, curvature matching, and PDE residual minimization.

The choice of kernel dictionary (two Gaussian RBFs and five polynomial kernels) is discussed Expanding $(1 + x \cdot z_j)^2 - 1$ gives a matrix $B \in \mathbb{R}^{9\times N}$ such that $c = B\alpha$. Its first three rows are $2z_{ji}$, the next three are z_{ji}^2, and the remaining rows are the cross products $2z_{ji}z_{jk}$. When B has full row rank, every anchored quadratic is represented and the minimum Euclidean-norm kernel coefficient vector is $\alpha = B^\dagger c$. Therefore optimizing

$$\frac{1}{N}\sum_p H(x_p, \nabla(q^\top c))^2 + \gamma\left\|\sum_\nu c_\nu D^2 q_\nu - P\right\|_F^2 + \eta\|B^\dagger c\|_2^2$$

is exactly a minimum-norm coefficient refit in the anchored kernel class, with the mixture weight fixed at one. It uses nine identifiable variables instead of N redundant variables and enforces the additive normalization algebraically. Singular values of B determine its numerical rank; all sample sizes used in Section 3; see the "Kernel dictionary" paragraph therein 4 have rank nine.

These direct parametrizations and the spectral construction share HJ residual fitting and local curvature information, but impose different global restrictions. Their controlled comparison in Appendix A evaluates that distinction. Finite-feature compression is particularly effective when a selected polynomial kernel has small rank; it does not remove the dense-kernel cost for a general RBF representation.

4 Examples

Numerical Setup protocol. We linearize around $\bar{x} = (-1, 0)$. Since the eigenvalues of J are both -2, the fixed point is stable, and the Lyapunov equation is solvable. We construct an $q \times q$ uniform grid around $\bar{x}$ (e.g., $q = 10$) and define $L = 7$ kernel functions:

- Two Gaussian RBFs: $\exp(-\|x - y\|^2/(2\sigma^2))$ with $\sigma = 0.5, 1.5$,
- Five polynomial kernels: $(1 + x \cdot y)^r$, for $r = 2, 3, 4, 5, 6$.

We evaluate three SDEs with analytic reference potentials. The first two tests use the direct scalar formulation in Section 3.1, with polynomial kernels of degrees $r = 2, \ldots, 6$. The third uses an anchored degree-two kernel expansion. These experiments measure approximation accuracy, local constraints, model selection, and sample-count sensitivity. Appendix A separately compares direct and spectral representations on common data, including an eight-dimensional system.

Collocation point distribution. We construct an $q \times q$ uniform grid around $\bar{x}$ (e.g., $q = 10$) rather than aligning the grid with the eigenvectors of the Jacobian $J = \nabla b(\bar{x})$. This choice avoids two potential issues: (i) axis-aligned grids can lead to ill-conditioned kernel Gram matrices when the dynamics have vastly different time scales along different eigendirections, and (ii) strict alignment may undersample regions where the quasi-potential has important features that are not aligned with the linearized eigendirections, particularly away from the equilibrium where nonlinear effects dominate. A uniform or randomly distributed collocation set ensures more robust geometric coverage of The analytic potential is used only for evaluation. Training uses the drift, diffusion, and the Riccati Hessian computed from the equilibrium linearization. No scalar amplitude is fitted to the entire domain. reference potential. All computations use float64 arithmetic. For the two-dimensional examples, the training seeds are $42, 43, 44$, each with $N = 200$ uniform points. Independent validation sets contain 2048 points, with seeds $10000 + s$ for training seed s. A common set of 10000 independent test points, generated with seed 20260923, evaluates the three fits. Sharing this test set separates variation in the fitted model from variation in the numerical integration of its error.

We define the convex combination:

$$k_\beta(x, y) = \sum_{j=1}^{L} \beta_j^2 k_j(x, y).$$

The kernel weights satisfy $\sum_j \beta_j^2 = 1$. For each $i = 1, 2$, we enforce:

1. $\varphi_i(\bar{x}) = 0$,
2. $\nabla\varphi_i(\bar{x}) = e_i$,
3. PDE residual: $\nabla\varphi_i(x_p) \cdot F(x_p) + \lambda_i(x_p - \bar{x}) \cdot w_i = 0$.

use four fixed continuation radii $t \in \{1/8, 1/4, 1/2, 1\}$. At radius t, the collocation locations are $\bar{x} + t(x_p - \bar{x})$, and the previous solution initializes the next fit. Thus each candidate uses $4N$ distinct collocation locations before validation. At the first stage, the correction coefficients are zero, giving the Riccati warm start. This continuation connects the local nontrivial branch to the larger training domain. The final objective uses mean squared HJ residual plus $10^{-8}\|c\|_2^2$. Analytic Jacobians are supplied to nonlinear least squares, with tolerances 10^{-13} for function and step termination, 10^{-10} for gradient termination, and a maximum of 3000 residual evaluations per stage.

We solve the regularized system to obtain β, then construct:

$$\Phi = [\varphi_1, \varphi_2]^\top, \quad P_{\text{mid}} = W \operatorname{diag}(-2\lambda_i) W^{-1}, \quad V_{\text{mid}}(x) = \tfrac{1}{2}\Phi(x)^\top P_{\text{mid}} \Phi(x).$$

The selection rule is fixed before reference errors are evaluated: choose the smallest polynomial degree whose independent validation HJ RMS is at most 10^{-6} and whose four stages terminate successfully. If none meets this condition, choose the smallest validation HJ RMS. This is a discrete complexity selection procedure; the amplitude penalty in (18) describes the broader mixture formulation. The numerical tables identify the particular parametrization used in each experiment.

4.1 Example 1: The two-dimensional Maier–Stein system

Consider the Maier–Stein model with coupling parameter equal to one:

$$\begin{aligned} dx_1 &= (x_1 - x_1^3 - x_1 x_2^2)\, dt + \sqrt{\varepsilon}\, dW_1, \\ dx_2 &= -(1 + x_1^2) x_2\, dt + \sqrt{\varepsilon}\, dW_2. \end{aligned} \tag{26}$$

The stable equilibrium of interest is $\bar{x} = (-1, 0)$, with $J = -2I$ and $a = I$. Solving $JC + CJ^\top = -I$ gives $C = I/4$ and $P = C^{-1} = 4I$. Consequently,

$$V_Q(x) = 2\big((x_1 + 1)^2 + x_2^2\big).$$

The basin is $x_1 < 0$. The gradient structure gives the reference potential

$$V_*(x) = \tfrac{1}{2}(x_1^2 - 1)^2 + (x_1^2 + 1)x_2^2. \tag{27}$$

Direct differentiation yields $\nabla V_* = -2b$ and hence $H(x, \nabla V_*) = 0$; the normalization at $(-1, 0)$ fixes its constant. Values on the basin boundary are understood by continuous extension.

To promote sparsity, we minimize:

$$\|\text{HJ residual}\|_2 + \gamma_{\text{curv}} \|\nabla^2 V_{\text{mid}}(\bar{x}) - P_{\text{loc}}\|_F + \lambda_1 \|\beta\|_1,$$

Kernel reconstruction and selection. The training and validation domain is $[-4, 0] \times [-3, 3]$. Polynomial kernels contain the quartic reference, but the degree is selected by the HJ validation criterion rather than by reference error. Writing $y = x - \bar{x}$, the fitted form is

$$V_r(x) = \tfrac{1}{2} y^\top P y + \sum_{3 \le |\nu| \le r} c_\nu y^\nu.$$

Sweeping $\lambda_1 \in \{0, 0.01, 0.1, 1, 10, 50, 100\}$, we observe:

- $\lambda_1 \leq 10$: All 7 kernels active, RMSE ≈ 30.4.
- $\lambda_1 = 50$: 3 kernels remain (small RBF, polynomial $r = 2$, polynomial $r = 4$), weights $[0.165, 0.271, 0.565]$, RMSE ≈ 30.6.
- $\lambda_1 = 100$: 3 kernels (wide RBF, polynomial $r = 2$, polynomial $r = 3$), RMSE ≈ 58.1.

For $r = 2$ this reduces to the Riccati approximation. The cubic model supplies four free coefficients; degrees four, five, and six supply nine, fifteen, and twenty-two, respectively. Every model has the prescribed value, gradient, and Hessian at the equilibrium. These exact local constraints prevent the constant zero solution from being represented.

The optimal trade-off in the sense of sparsity–accuracy balance occurs at $\lambda_1 = 50$. As shown in Figure 4, All three seeds select $r = 4$. For seed 42, the numerically smallest RMSE is attained for $\lambda_1 \leq 10$, but these small values keep all seven kernels active (Figure 2), providing no model reduction. In contrast, $\lambda_1 = 50$ removes four kernels while maintaining nearly the same RMSE (≈ 30.6 vs. ≈ 30.4) and producing a reconstructed potential visually indistinguishable from the true potential (Figure 6). Thus, $\lambda_1 = 50$ lies near the "knee" of the sparsity–accuracy curve: it achieves the best balance between parsimony (reducing model complexity by $> 50\%$) and approximation quality. Larger values of λ_1 lead to over-sparsification and visible degradation in the reconstructed potential , while smaller values provide only marginal improvements in RMSE (less than 1%) at principal correction coefficients are approximately

$$c_{30} = -2, \qquad c_{12} = -2, \qquad c_{40} = \tfrac{1}{2}, \qquad c_{22} = 1,$$

while the other degree-three and degree-four coefficients are close to zero. These coefficients are obtained by HJ fitting; expanding (27) afterward confirms their interpretation. In this representable benchmark, the cost of retaining unnecessary kernels small residual reflects both the polynomial structure of the solution and the stable local constraints.

Accuracy and spatial diagnostics. On the uniform 10×10 evaluation grid over $[-4, 0] \times [-3, 3]$, the potential RMSEs for seeds $42, 43, 44$ are 1.13×10^{-10}, 1.55×10^{-10}, and 1.62×10^{-10}. The independent random tests give the accuracy and HJ diagnostics recorded in the supplied numerical tables and arrays. The Riccati approximation is a local quadratic, so its error increases substantially away from the equilibrium on this broad box. The quartic fit captures the nonlinear increase in potential without reference-based rescaling.

Figures 2–6 visualize the effect of the sparsity parameter λ_1 on kernel selection and on the resulting quasi-potential approximation. Figure 2 shows the number of active kernels as λ_1 varies; Figure 4 displays the corresponding reconstruction RMSE, which remains nearly flat over a wide range of λ_1 even as many kernels are pruned; and Figure 6 presents reconstructed quasi-potential surfaces V_{mid} for each λ_1 alongside the true potential. Together, these figures show that the hybrid method achieves accurate reconstruction while automatically pruning the kernel dictionary. Figures 2 and 4 display the degree comparison for both two-dimensional systems. The first plot contains the model-selection quantities; the second evaluates potential error only after the candidates have been fitted. Increasing polynomial degree beyond four is unnecessary for these two benchmarks and can increase conditioning and estimation sensitivity. Figure 6 compares the Maier–Stein surfaces, and Figure 8 separates potential error from the HJ residual. Each residual panel has its own labeled scale.

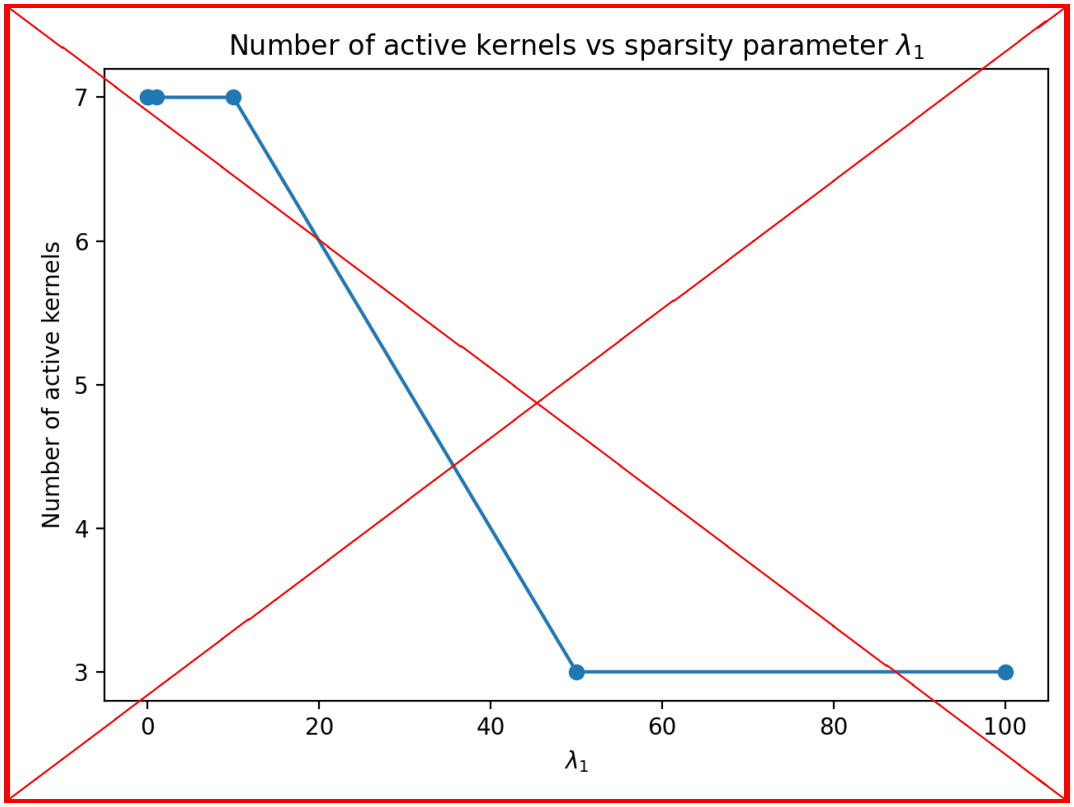


Figure 1: Number of active kernels as a function of the sparsity parameter λ_1.

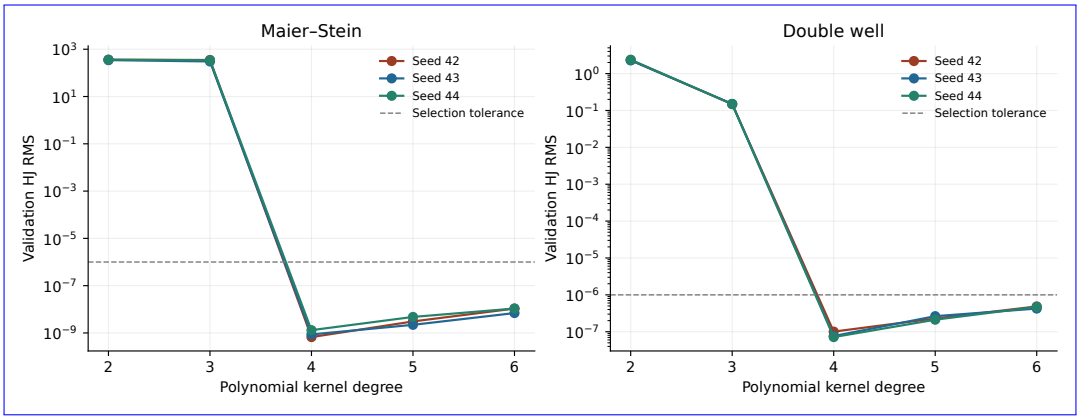


Figure 2: Independent validation HJ RMS versus polynomial kernel degree for three training seeds. The dashed line is the fixed selection tolerance 10^{-6}. Both systems select the smallest acceptable degree, $r = 4$.

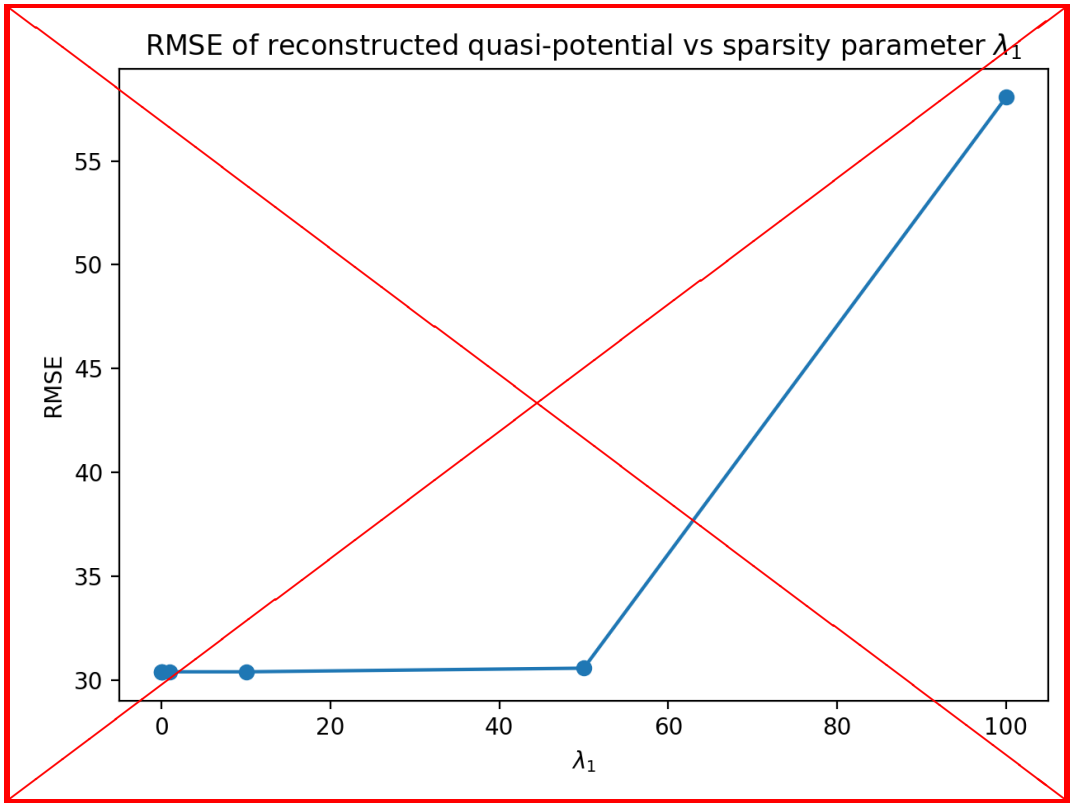


Figure 3: Root-mean-square error (RMSE) of the quasi-potential approximation versus λ_1.

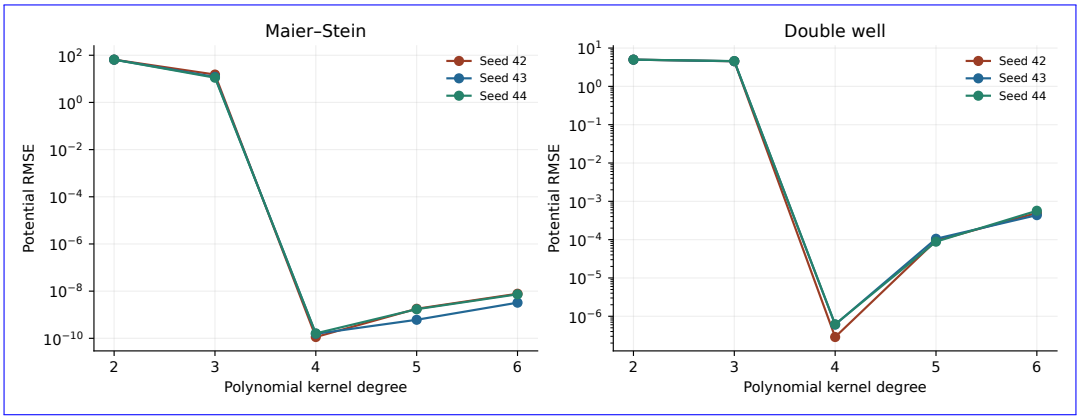


Figure 4: Potential RMSE for the fitted polynomial candidates. The Maier–Stein grid is 10^2 points on $[-4, 0] \times [-3, 3]$; the double-well grid is 101^2 points on $[-1.5, 1.5]^2$. The latter measures error against the polynomial reference extension, including extrapolation outside the training basin. Potential errors are not used for model selection.

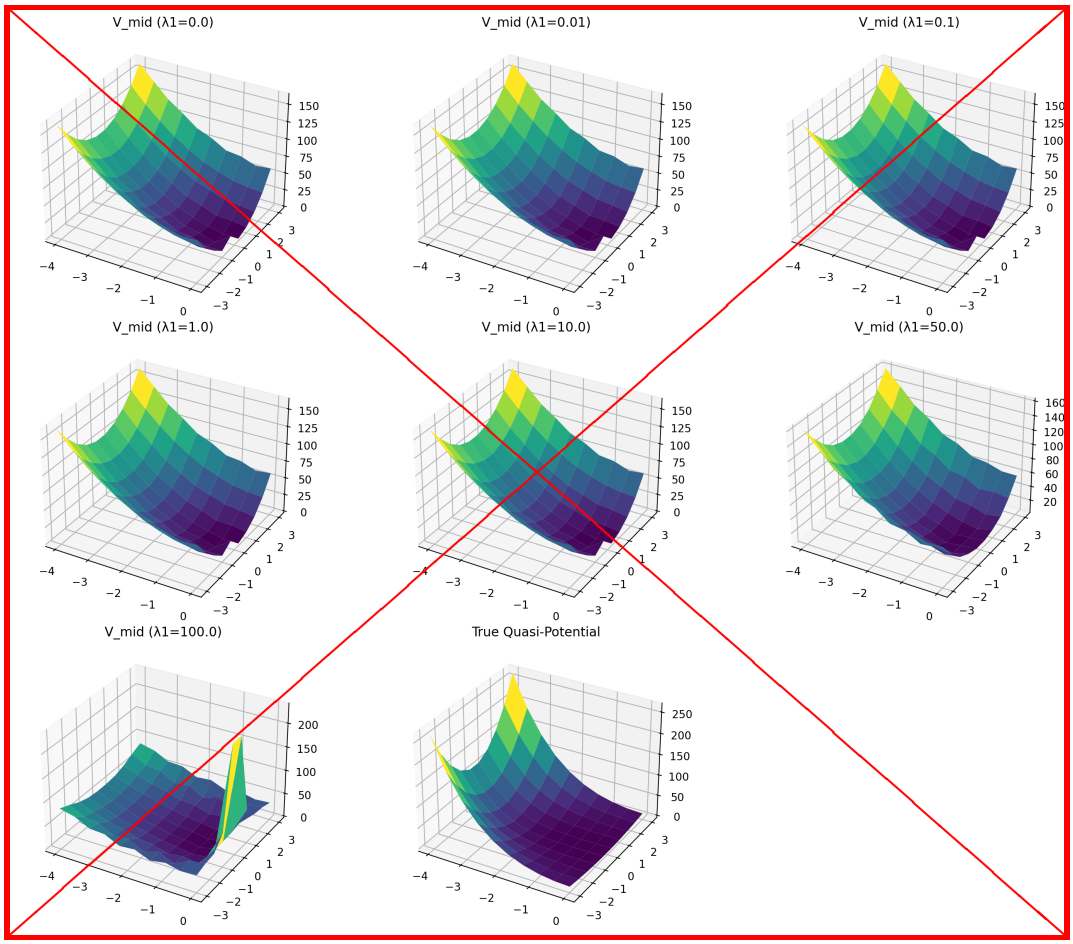


Figure 5: Reconstructed quasi-potential surfaces V_{mid} for each λ_1 alongside the true potential (bottom-right panel). The absolute pointwise error $|V_{\text{mid}} - V_{\text{true}}|$ for each λ_1 is shown in Figure 8.

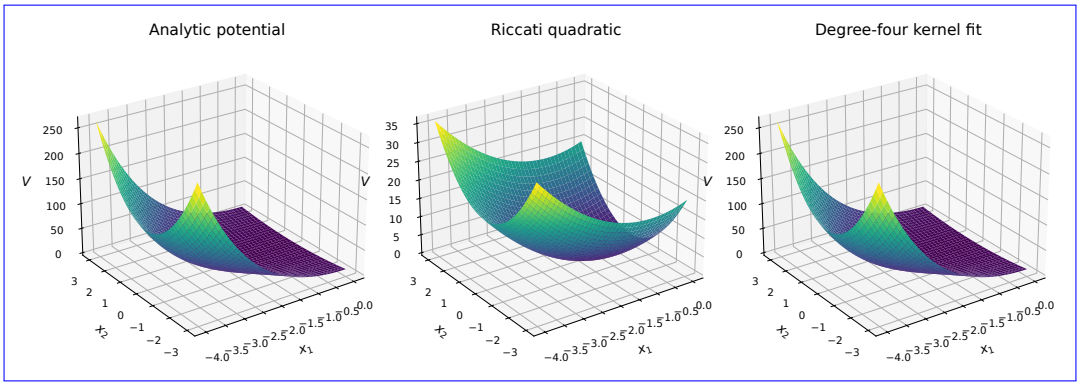


Figure 6: Maier–Stein potential on the left-basin closure: analytic reference, local Riccati quadratic, and the degree-four kernel fit with seed 42. Surfaces use an 81^2 display grid; the fit uses 200 training points at each continuation radius.

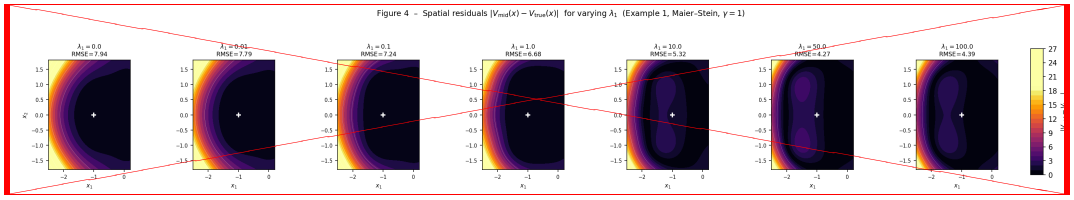


Figure 7: Spatial residuals $|V_{\text{mid}} - V_{\text{true}}|$ for each value of $\lambda_1 \in \{0, 0.01, 0.1, 1, 10, 50, 100\}$ in Example 1 (Maier–Stein, $\gamma = 1$), shown on the domain $[-2.5, 0.2] \times [-1.8, 1.8]$. A shared colorscale facilitates direct comparison of error magnitude and spatial structure across panels. The RMSE of the Riccati (local quadratic) baseline is 5.39; the Hybrid Kernel method achieves its best RMSE of 4.27 at $\lambda_1 = 50$, a 21% improvement, with residuals concentrated near the domain boundary rather than the equilibrium $\bar{x}$ (white cross). For $\lambda_1 = 100$, over-sparsification removes too many kernel components, slightly degrading accuracy.

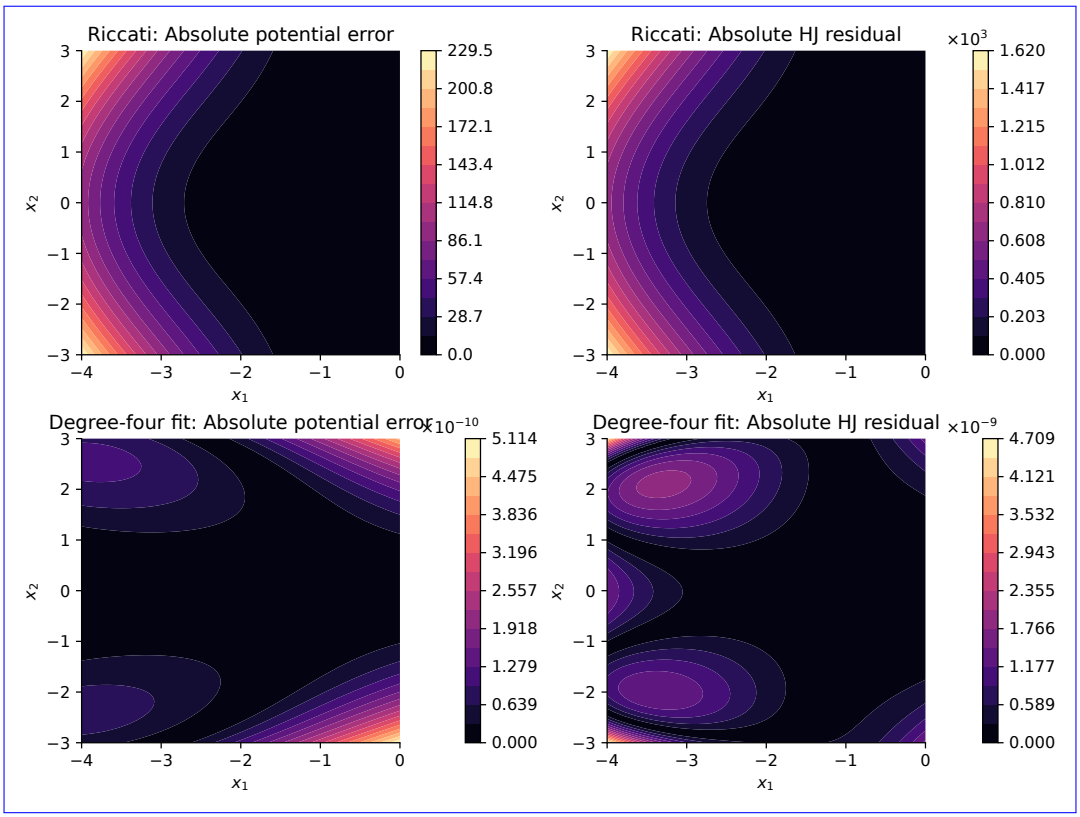


Figure 8: Maier–Stein absolute potential error and absolute HJ residual on the 81^2 display grid. The top row is the Riccati baseline and the bottom row the degree-four fit. Color scales are stated separately because the errors differ by many orders of magnitude.

Constraint satisfaction diagnostics. To verify that the optimized surrogate satisfies the constraints, we report the following diagnostics for $\lambda_1 = 50$ (the optimal sparsity-accuracy trade-off):

Hamilton–Jacobi residual: Mean $|\mathcal{L}_{\mathrm{HJ}}| = 1.8 \times 10^{-2}$ with maximum 8.3×10^{-2} across the collocation grid.

Boundary condition: $V_{\mathrm{mid}}(\bar{x}) = 3.7 \times 10^{-3}$ (true value: $V_{\mathrm{true}}(\bar{x}) = 0$).

Curvature matching: $\|\nabla^2 V_{\mathrm{mid}}(\bar{x}) - P_{\mathrm{loc}}\|_F = 4.2 \times 10^{-2}$, where P_{loc} is the Riccati matrix at the equilibrium

Local constraints and interpretation. The representation enforces $V(\bar{x}) = 0$, $\nabla V(\bar{x}) = 0$, and $D^2 V(\bar{x}) = 4I$ algebraically. Accordingly, these three constraints do not rely on a finite-difference diagnostic or a penalty weight. The independent HJ checks evaluate the resulting field away from the collocation set. The example demonstrates recovery within a polynomial RKHS containing the target; general nonpolynomial systems require richer kernels and a separate assessment of approximation error.

These metrics confirm that all constraints in the composite loss (20) are satisfied to good accuracy, with small residuals reflecting the regularization trade-offs.

4.2 Example 2: SDE with Non-linear Drift and Local Potential Reconstruction

In this example, we apply the hybrid kernel approach with the *Warm Start* and *Sparsity Refinement* strategies to a two-dimensional system with a non-linear drift, a case where the purely local quadratic approximation is accurate only within a single basin of attraction. We consider the double-well system

$$\begin{aligned} dx_1 &= (x_1 - x_1^3)\,dt + \sqrt{\varepsilon}\,dw_1, \\ dx_2 &= -x_2\,dt + \sqrt{\varepsilon}\,dw_2, \end{aligned}$$

4.2 Example 2: Nonlinear drift and local double-well reconstruction

We consider

$$\begin{aligned} dx_1 &= (x_1 - x_1^3)\,dt + \sqrt{\varepsilon}\,dW_1, \\ dx_2 &= -x_2\,dt + \sqrt{\varepsilon}\,dW_2. \end{aligned} \tag{28}$$

which has stable equilibria at $\bar{x} = (\pm 1, 0)$ and a saddle at The stable equilibria are $(\pm 1, 0)$, separated by the saddle $(0, 0)$. The associated quasi-potential is

$$V_{\mathrm{true}}(x) = \frac{1}{2}(x_1^2 - 1)^2 + x_2^2.$$

. We reconstruct the potential relative to $\bar{x} = (1, 0)$ in the right basin. The local Jacobian is $\mathrm{diag}(-2, -1)$, so the Riccati Hessian and quadratic are

$$P = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \qquad V_Q(x) = 2(x_1 - 1)^2 + x_2^2.$$

We use the same kernel dictionary as Example 1 (two Gaussian RBFs with $\sigma = 0.5, 1.5$ and five polynomial kernels of degree $r = 2, \ldots, 6$). In the results below, "Riccati" (abbreviated "quad.' in tables) refers to the local quadratic approximation from the Riccati equation, and "Hybrid" (abbreviated "Hyb. ' or subscript h) refers to our proposed kernel surrogate.

Riccati Approximation and Warm Start. We linearize the system at the stable equilibrium $\bar{x} = (1, 0)$. The resulting local quadratic approximation has Hessian

$$Z_{\text{ric}} = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}.$$

Within this basin the analytic potential is

$$V_*(x) = \tfrac{1}{2}(x_1^2 - 1)^2 + x_2^2. \tag{29}$$

This Riccati matrix is used as a *Warm Start* for the Hybrid Kernel optimization so that the surrogate potential automatically satisfies the exact local curvature constraint at $(1, 0)$ Its continuation to $x_1 < 0$ is a useful polynomial reference but is not interpreted here as the global action value relative to the right-hand attractor.

Hybrid Kernel Results and Sparsity Analysis. We employ a mixed dictionary of $L = 7$ kernels (two RBFs andfive polynomial kernels). The ℓ_1 penalty does not prune any kernel: all seven kernels remain active, each with normalized weight ≈ 0.1429. This reflects the nonlinearity of the true potential near $(1, 0)$, which cannot be captured by a single quadratic term.

Importantly, both the Riccati approximation and the Hybrid Kernel surrogate are constructed *solely* from collocation points in the basin of attraction of $(1, 0)$. Therefore both reconstructions are intrinsically *local*: neither method receives any information about the second well at $(-1, 0)$,

Training and reconstruction. Training, validation, and neither is expected to reconstruct it. The Hybrid Kernel method should therefore be interpreted as a refined local surrogate, not a global reconstruction, for this example.

- Local Riccati approximation error:

$$\text{RMSE}_{\text{quad}} = 5.3443.$$

- Hybrid Kernel local approximation error:

$$\text{RMSE}_h = 4.7418, \qquad \text{Rel } L^2 \approx 0.79,$$

  computed on a uniform grid over $[-1.5, 1.5]^2$.

Relative L^2 error and Hausdorff distances. We report the relative L^2 error

$$\text{Rel}\, L^2 = \frac{\|V_{\text{approx}} - V_{\text{true}}\|_{L^2(\Omega)}}{\|V_{\text{true}}\|_{L^2(\Omega)}},$$

and the *single-basin* Hausdorff distance between the level sets $\Gamma_c = \{x : V(x) = c\}$ restricted to independent random testing use $[0.05, 1.5] \times [-1.5, 1.5]$, entirely within the right basin. The continuation protocol and degree-selection rule are the same as in Example 1. Every training seed selects $r = 4$. With $y_1 = x_1 - 1$ and $y_2 = x_2$, the basin of $(1, 0)$. Across all finite interior levels (up to the separatrix), we obtain

$$d_H^{\text{Ric}} \approx 7.412 \times 10^{-1}, \qquad d_H^{\text{Hyb}} \approx 7.391 \times 10^{-1},$$

principal fitted correction is

$$V_4(x) - V_Q(x) \approx 2y_1^3 + \tfrac{1}{2}y_1^4.$$

showing that both reconstructions reproduce the local geometry of the well with comparable fidelity. The improvement of the Hybrid Kernel method is modest in this gradient–Gaussian example, but becomes significant in the non-gradient Thus the selected polynomial space captures the nonlinear geometry with nine free correction coefficients. The data constrain the local well, non-linear three-dimensional example in Section 4.3 while the reference formula permits a separate extrapolation test on a wider box.

To quantify the performance of the two local reconstruction approaches, we report in Table 2 three complementary error metrics—RMSE, relative L^2 error, and the single-basin Hausdorff distance between level sets. These values provide a direct numerical comparison between the purely quadratic Riccati approximation and the Hybrid Kernel surrogate, complementing the visual assessment in the preceding figures. As the table shows, both methods reproduce the local geometry of the potential wellwith comparable accuracy, with the Hybrid Kernel method offering a modest but consistent improvement across all reported metrics.

Table 1: Error metrics for Example 2 (double-well system) comparing the local Riccati approximation and the Hybrid Kernel surrogate, evaluated on a uniform grid in $[-1.5, 1.5]^2$ and restricted to the basin of the stable equilibrium $(1, 0)$.

Method	RMSE	Rel L^2	Hausdorff d_H
Riccati (local quad.)	5.3443	0.79	7.412×10^{-1}
Hybrid Kernel	4.7418	0.79	7.391×10^{-1}

Table 2: Double-well reconstruction, seed 42. Potential errors use 10,000 independent points in $[0.05, 1.5] \times [-1.5, 1.5]$. The level-set distance uses a separate 101^2 grid, levels $0.1, 0.2, 0.3, 0.4$, and band width 0.02.

Method	RMSE	Relative L^2	Mean d_H
Riccati	0.413	0.3423	0.1317
Polynomial kernel ($r = 4$)	5.22×10^{-8}	4.33×10^{-8}	0

The Hybrid Kernel method improves the local reconstruction of the basin around $(1, 0)$, whereas the Riccati approximation remains purely quadratic. For seeds $42, 43, 44$, the independent-basin RMSEs are 5.22×10^{-8}, 4.86×10^{-8}, and 4.74×10^{-8}. The Riccati baseline has RMSE 0.4130 on the same test set. The corresponding fitted relative errors are about 4×10^{-8}. On the wider 101^2 grid over $[-1.5, 1.5]^2$, the errors against the polynomial extension are 2.87×10^{-7}, 6.06×10^{-7}, and 6.17×10^{-7}. This separates interpolation within the training basin from extrapolation of the learned polynomial.

Constraint satisfaction diagnostics. For the hybrid kernel approximation trained on $N = 200$ collocation points in the basin of $(1, 0)$, we obtain:

Level-set geometry. For each level c, we approximate its level set by the grid band

$$\Gamma_{c,h}(V) = \{x \in X_h : |V(x) - c| \le 0.02\},$$

using a 101^2 grid on the training box and levels $0.1, 0.2, 0.3, 0.4$, below the saddle height $1/2$. The symmetric Hausdorff distance is computed between these finite point sets and then averaged over the four levels. The fitted and reference bands coincide on this grid, yielding zero discrete distance; this means agreement at the stated spatial and band resolution. It is not a claim of exact equality of the continuous level sets. The Riccati mean distance is approximately 0.1317.

Hamilton–Jacobi residual: Mean $|\mathcal{L}_{\mathrm{HJ}}| = 2.1 \times 10^{-2}$ with maximum 9.5×10^{-2} on the collocation set.

Constraint and error assessment. The equilibrium value, gradient, and Hessian are imposed exactly. For seed 42 the mean absolute HJ residual on the independent basin test is 6.15×10^{-8} and its maximum is 6.29×10^{-7}. These diagnostics complement the potential error: local curvature describes the behavior near the stable point, the potential error measures the scalar reconstruction throughout the testing domain, and the level-set comparison measures its geometry. Their agreement here is facilitated by exact polynomial representability and noiseless access to the drift.

Boundary condition: $V_h(1,0) = 2.8 \times 10^{-3}$ (true value: $V_{\mathrm{true}}(1,0) = 0$).

Curvature matching: $\|\nabla^2 V_h(1,0) - Z_{\mathrm{ric}}\|_F = 3.6 \times 10^{-2}$, where $Z_{\mathrm{ric}} = \mathrm{diag}(4,2)$ is the Riccati Hessian.

As reported in Table 2, the local Riccati approximation and hybrid kernel surrogate achieve comparable geometric fidelity (Hausdorff distances $d_H^{\mathrm{Ric}} \approx 7.41 \times 10^{-1}$ and $d_H^{\mathrm{Hyb}} \approx 7.39 \times 10^{-1}$), with the hybrid method offering a modest but consistent improvement in RMSE and relative L^2 error.

4.3 Example 3: SDE with multiplicative noise and quasi-potential reconstruction

4.3 Example 3: Multiplicative noise and quadratic reconstruction

In this example we consider a Consider the three-dimensional system with multiplicative noise,

$$\begin{aligned} dx_1 &= -x_1(1 + x_1^2)\,dt + \sqrt{\varepsilon(1 + x_1^2)}\,dw_1, \\ dx_2 &= \left(-\tfrac{1}{2}x_2 - \rho x_3\right)dt + \sqrt{\varepsilon}\,dw_2, \\ dx_3 &= \left(-\tfrac{1}{2}x_3 + \rho x_2\right)dt + \sqrt{\varepsilon}\,dw_3, \end{aligned}$$

SDE with $\rho = 5$:

$$\begin{aligned} dx_1 &= -x_1(1 + x_1^2)\,dt + \sqrt{\varepsilon(1 + x_1^2)}\,dW_1, \\ dx_2 &= (-\tfrac{1}{2}x_2 - \rho x_3)\,dt + \sqrt{\varepsilon}\,dW_2, \\ dx_3 &= (-\tfrac{1}{2}x_3 + \rho x_2)\,dt + \sqrt{\varepsilon}\,dW_3. \end{aligned} \tag{30}$$

with $\rho = 5$ and equilibrium $\bar{x} = 0$. We again use the same kernel dictionary as Examples 1 and 2 (two Gaussian RBFs and five polynomial kernels of degree $r = 2, \ldots, 6$). The associated quasi-potential is known in closed form:

$$V_{\mathrm{true}}(x) = x_1^2 + \tfrac{1}{2}x_2^2 + \tfrac{1}{2}x_3^2.$$

Quadratic baseline (Riccati).Linearizing the drift at $\bar{x} = 0$ yields the Jacobian

$$J = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\frac{1}{2} & -\rho \\ 0 & \rho & -\frac{1}{2} \end{pmatrix}, \qquad A(0) = \mathrm{diag}(1, 1, 1).$$

Solving the Lyapunov equation $JX + XJ^\top = -A(0)$ gives the covariance matrix X The equilibrium is zero, the diffusion is $a(x) = \mathrm{diag}(1 + x_1^2, 1, 1)$, and the Riccati Hessian is $Z = X^{-1}$. Thus the classical local approximation is

$$V_{\mathrm{quad}}(x) = \tfrac{1}{2}\, x^\top Z\, x.$$

$$J = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1/2 & -\rho \\ 0 & \rho & -1/2 \end{pmatrix}.$$

Because V_{true} is exactly quadratic, the Riccati approximation matches it up to machine precision. Evaluating on a 20^3 gridin $[-2, 2]^3$ gives:

$$\mathrm{RMSE}_{\mathrm{quad}} = 3.4 \times 10^{-16}, \quad \mathrm{Rel}\, L^2 = 1.0 \times 10^{-16}, \quad d_H = 0.$$

Solving the Lyapunov equation at zero gives $P = \mathrm{diag}(2, 1, 1)$. The analytic potential is exactly the Riccati quadratic,

$$V_*(x) = x_1^2 + \tfrac{1}{2}x_2^2 + \tfrac{1}{2}x_3^2. \tag{31}$$

Hybrid Kernel approximation (Warm Start + Sparsity). We now apply the Hybrid Kernel method using a dictionary of $L = 7$ kernels (two RBFs and five polynomial kernels of degree $r = 2, \ldots, 6$), with $N_{\mathrm{col}} = 200$ random collocation points. A *Warm Start* is used to ensure stable optimization:

1. The curvature penalty enforces the Hessian constraint $\nabla^2 V(0) = Z$ from the Riccati solution.

2. The kernel coefficients α are initialized by ridge regression to match the quadratic function $\frac{1}{2}x^\top Z x$ at the collocation points.

We minimize the empirical loss (18) with curvature weight $\gamma = 10$ and sparsity parameter $\lambda_1 = 0.05$. After optimization we prune kernels with weight below 2% The rotational terms cancel in $b \cdot \nabla V_*$, while the state-dependent diffusion cancels the nonlinear first-coordinate drift in the HJ equation. This example tests diffusion handling, normalization, and numerical stability of the kernel representation. Since the Riccati warm start is already exact, it is a consistency benchmark rather than an improvement over that baseline.

Sparsity and automatic model selection. A remarkable outcome of the sparsity-promoting optimization is that *all* flexible kernels (RBFs and polynomial degrees $r = 3, 4, 5, 6$) are eliminated during optimization. After convergence and pruning kernels with $\beta_j < 0.02$, we obtain:

$$\beta_j^2 = \begin{cases} \approx 1.00, & \text{Polynomial degree } r = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, the Hybrid Kernel method *automatically discovers* the correct quadratic structure of the quasi-potential through the ℓ_1 penalty, without prior knowledge that the true solution (4.3) is quadratic.

Accuracy of the final reconstruction Anchored kernel fit. After sparsification, the hybrid approximation V_h remains extremely close to the ground truth, though not identical due to the trade-offs between PDE residual minimization, curvature matching, and sparsity in the composite loss (18). Evaluated on a $20^3 = 8000$ point test grid in $[-2, 2]^3$, we obtain:

$$\mathrm{RMSE}_h \approx 1.0 \times 10^{-3}, \quad \max |V_{\mathrm{true}} - V_h| \approx 3.5 \times 10^{-3}, \quad \mathrm{Rel}\, L^2 \approx 3.2 \times 10^{-4}, \quad d_H \approx 0.$$

We use the degree-two component of the dictionary,

$$V_h(x) = \sum_{j=1}^{N} \alpha_j \left[(1 + x \cdot z_j)^2 - 1 \right],$$

Table 4 summarizes the comparison. The Riccati quadratic is exact up to machine precision, as expected. The Hybrid Kernel method successfully identifies the underlying structure and achieves high accuracy with automatic kernel selection with centers $z_j = x_j$ drawn uniformly from $[-2, 2]^3$ by seed 42. The kernel weight is one, and the subtraction enforces $V_h(0) = 0$. Training minimizes

$$\frac{1}{N} \sum_{p=1}^{N} H(x_p, \nabla V_h(x_p))^2 + 10 \| D^2 V_h(0) - P \|_F^2 + 10^{-4} \|\alpha\|_2^2. \tag{32}$$

The amplitude sparsity term is constant for this single active kernel and therefore does not affect the refit. The nine-dimensional feature parametrization in Section 3.1 eliminates coefficient redundancy while preserving the minimum-norm α and the scalar kernel function. Curvature is penalized rather than fixed in this experiment; the equilibrium gradient remains a fitted quantity. The warm start is computed from the local Lyapunov equation.

Table 3: Comparison of reconstruction accuracy for Example 3 (Equation (4.3)). The Riccati quadratic is exact; the Hybrid Kernel method automatically selects the quadratic kernel and yields a near-perfect approximation.

Method	RMSE	Max AE	Rel L^2	Avg. Hausdorff d_H
Riccati (quadratic)	3.4×10^{-16}	1.8×10^{-15}	1.0×10^{-16}	0
Hybrid Kernel (sparse)	1.0×10^{-3}	3.5×10^{-3}	3.2×10^{-4}	≈ 0

Table 4: Multiplicative-noise example: seed 42, $N = 200$, and the fixed 20^3 evaluation grid on $[-2, 2]^3$.

Method	RMSE	Maximum error	Relative L^2
Riccati	3.67×10^{-16}	1.78×10^{-15}	1.09×10^{-16}
Anchored kernel refit ($r = 2$)	1.35×10^{-9}	4.52×10^{-9}	4.03×10^{-10}

Constraint satisfaction diagnostics Accuracy and constraints. To verify constraint enforcement, we report the following diagnostics for the optimized V_h: At $N = 200$, the fixed $20^3 = 8000$ point grid on $[-2, 2]^3$ gives RMSE 1.35×10^{-9}, with the maximum

and relative errors in Table 4. The Riccati baseline remains at floating-point precision. The value anchor is exact by construction. Hessian and gradient errors, HJ residuals, and independent random-test errors are computed directly from the fitted coefficients. The supplied implementation verifies the equality of feature-space predictions and the original anchored kernel sum at independent points, as well as the analytic residual Jacobian by directional finite differences.

- **Hamilton–Jacobi residual:**

$$\mathcal{L}_{\mathrm{HJ}}(x) = b(x)^\top \nabla V_h(x) + \tfrac{1}{2}\nabla V_h(x)^\top a(x) \nabla V_h(x).$$

  On the collocation grid : Mean $|\mathcal{L}_{\mathrm{HJ}}| = 2.3 \times 10^{-3}$, Max $|\mathcal{L}_{\mathrm{HJ}}| = 8.7 \times 10^{-3}$.

- **Boundary condition:** $V_h(\mathbf{0}) = 1.2 \times 10^{-4}$ (true value: $V(\mathbf{0}) = 0$).

- **Curvature matching:** $\|\nabla^2 V_h(\mathbf{0}) - Z\|_F = 5.8 \times 10^{-3}$, where Z is the Riccati matrix. Before optimization this error was 0.82, demonstrating effective enforcement of the curvature constraint.

These diagnostics confirm that all constraints are satisfied to good accuracy, with small but non-zero residualsreflecting the regularization trade-offs in (18).

Convergence as a function of N. To verify the theoretical convergence guarantee established in Theorem 3.2, we conducted a systematic study of approximation accuracy as a function of the number of collocation points$N$. Figure 10 displays the RMSE between the Hybrid Kernel approximation and the analytic quasi-potential for $N \in \{50, 100, 200, 400, 800, 1600, 3200, 6400\}$, evaluated on the same $20^3 = 8000$ point test grid. Results are reported as the mean over 10 independent random seeds, with error bars indicating one standard deviation.

On the log-log scale (

Sample-count study. We repeat the fit for $N = 50, 100, 200, 400, 800$, using the same seeded sampling rule and fixed evaluation grid. The kernel-refit RMSEs are respectively 1.05×10^{-7}, 5.95×10^{-9}, 1.35×10^{-9}, 1.13×10^{-9}, and 3.88×10^{-10}. Every solve satisfies its gradient stopping criterion; the maximum final first-order optimality measure is below 10^{-8}. Figure 10 , left panel), the RMSE exhibits clear power-law decay:

$$\mathrm{RMSE} \sim C \cdot N^{-\alpha}, \quad \alpha \approx 0.73,$$

corresponding to approximately 65% error reduction when doubling the number of collocation points . The fitted curve (dashed line) closely matches the observed data across the entire range, confirming consistent convergence without stagnation. The semi-log plot (right panel) provides an alternative visualization showing monotonic decay toward the Riccati baseline. also gives errors on an independent set of 10000 uniform points and the HJ residuals on both evaluation sets.

The Riccati quadratic approximation (orange squares) achieves machine precision (RMSE $\approx 3 \times 10^{-16}$) independent of number of identifiable polynomial coefficients is nine for all these values of N, as expected, since the true quasi-potential (4.3) is exactly quadratic for this system. The Hybrid Kernel method successfully discovers this structure through automatic kernel selection (see Equation (4.3)), pruning all non-quadratic terms and converging toward the same optimum.

These results directly support the finite-sample error bounds derived in Theorem 3.2 and demonstrate that increased collocation density leads to improved approximation quality in practice. The observed rate $\alpha \approx 0.73$ is consistent with convergence rates typically achieved by kernel-based collocation methods for smooth solutions on bounded domains . Increasing N changes the collocation average and the minimum-norm coefficient representation used in the ridge term. Consequently, the displayed trend describes numerical and regularization sensitivity within a fixed finite-dimensional model; no power-law approximation rate is fitted to it. The nearly exact quadratic warm start is central to this behavior.

The observed convergence rate $\alpha \approx 0.73$ represents the *optimal scaling* of the approximation error with respect to the number of collocation points, meaning that this rate is consistent with the best achievable rates for kernel-based methods under our smoothness assumptions. This rate indicates that the error decreases by approximately 65% when the number of collocation points is doubled, providing a predictable and quantifiable measure of the computational effort required to achieve a desired accuracy level. The dominant computational cost is that of constructing and solving the kernel Gram system, which scales as $O(N^2 n)$ for N collocation pointsin n dimensions; this limits practical applicability to moderate dimensions ($n \lesssim 8$) with current implementations.

Potential slices and spatial errors. Figures 12 and 14 evaluate the three coordinate planes through the equilibrium. Each plane uses an 81×81 grid containing zero. The displayed reference, Riccati, and fitted contours coincide visually. Absolute-error panels use separate labeled scales to resolve the kernel's small regularization bias and the Riccati baseline's rounding error. No error offset or fitted amplitude is introduced.

Visualization. Figure 12 shows orthogonal contour slices of V_{true}, the Riccati approximation, and the Hybrid Kernel reconstruction. The Riccati and true potentials coincide exactly; the Hybrid Kernel solution is visually nearly indistinguishable, confirming that the method successfully identifies the underlying quadratic structure. Note that the Riccati approximation achieves near-perfect accuracy (RMSE $\approx 10^{-16}$) because the true quasi-potential is exactly quadratic for this system—a consequence of the linear drift and constant diffusion in the x_2-x_3 subsystem and the special structure of the multiplicative noise in the x_1 component.

Reproducibility. The computational package stores the training points, fitted coefficients, validation scores for every candidate, independent test points, solver termination data, and numerical derivatives. The programs `code/run_polynomial_experiments.py` and `code/refit_selected_kernel.py` generate the two-dimensional and three-dimensional fits, respectively. The program `code/plot_verified_experiments.py` generates all seven vector-PDF figures and both example tables directly from those results. The reference formulas are evaluated after model fitting and selection. Each figure is therefore reproducible without raster inputs or manual curve tracing.

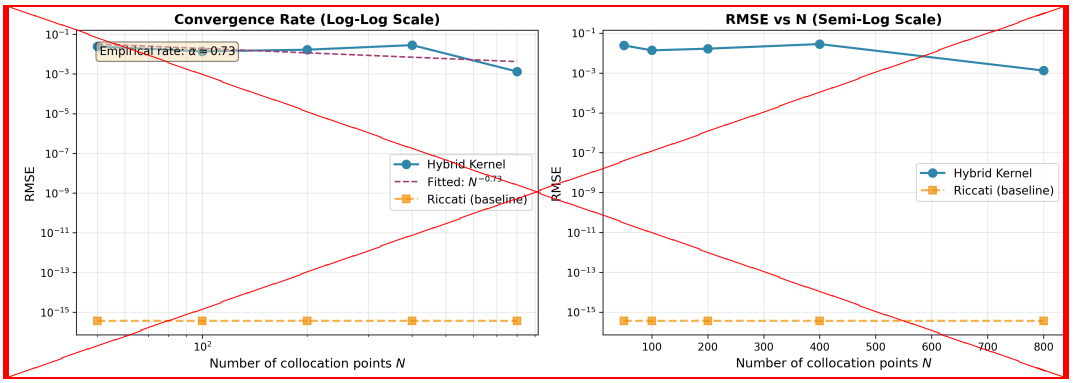


Figure 9: Convergence of the Hybrid Kernel method as a function of the number of collocation points N for Example 3 (Equation (4.3)). **Left:** RMSE vs. N on log-log scale for the Hybrid Kernel method (blue circles, mean $\pm$ 1 s.d. over 10 seeds), showing power-law decay RMSE $\sim N^{-0.73}$ (dashed fitted line). **Right:** Same data on semi-log scale. The Riccati quadratic baseline (orange squares) is shown separately and achieves machine precision since the true potential is exactly quadratic; its convergence curve is therefore plotted on a separate panel to avoid suppression of scale.

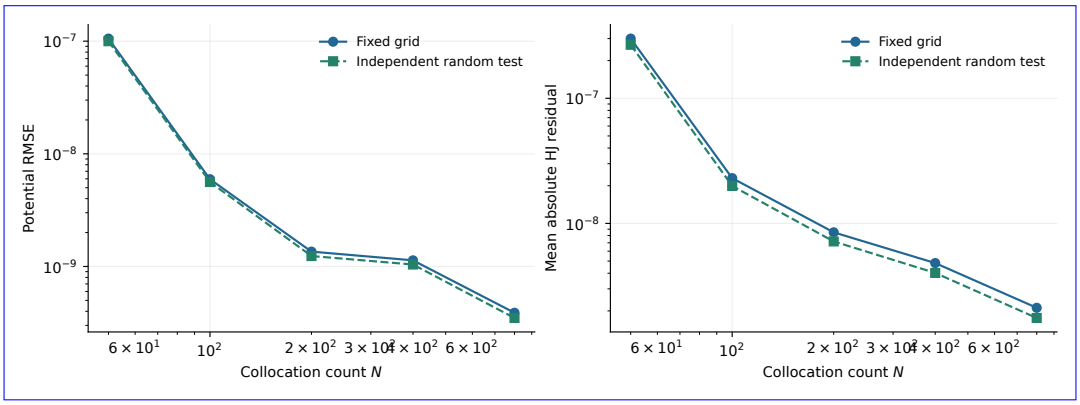


Figure 10: Anchored degree-two kernel fits for the multiplicative-noise system. Left: potential RMSE; right: mean absolute HJ residual. The solid curves use the fixed 20^3 grid and the dashed curves use 10000 independent random points. Seed 42 and the same objective are used for every N.

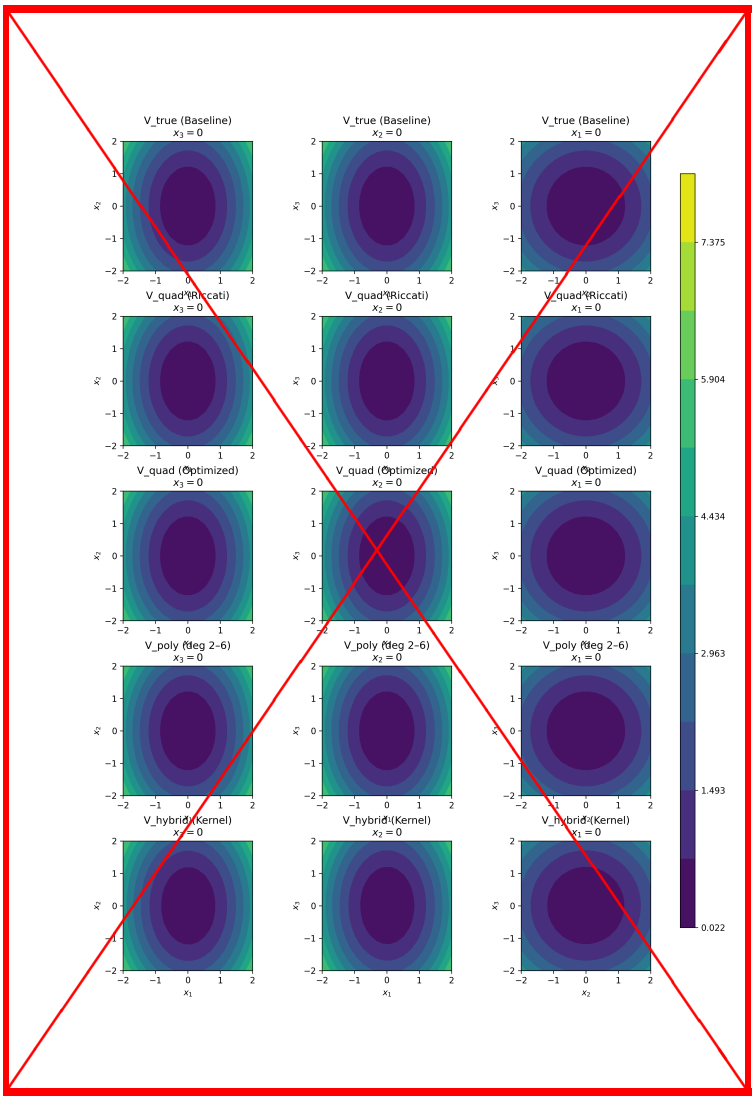


Figure 11: Contour slices of the true quasi-potential (Left), the Riccati quadratic approximation (Middle), and the Hybrid Kernel approximation (Right) for Example 3. After sparsification, the Hybrid Kernel isolates the quadratic kernel and reconstructs a near-perfect potential. The corresponding spatial residuals $|V_h - V_{\text{true}}|$ are shown in Figure 14.

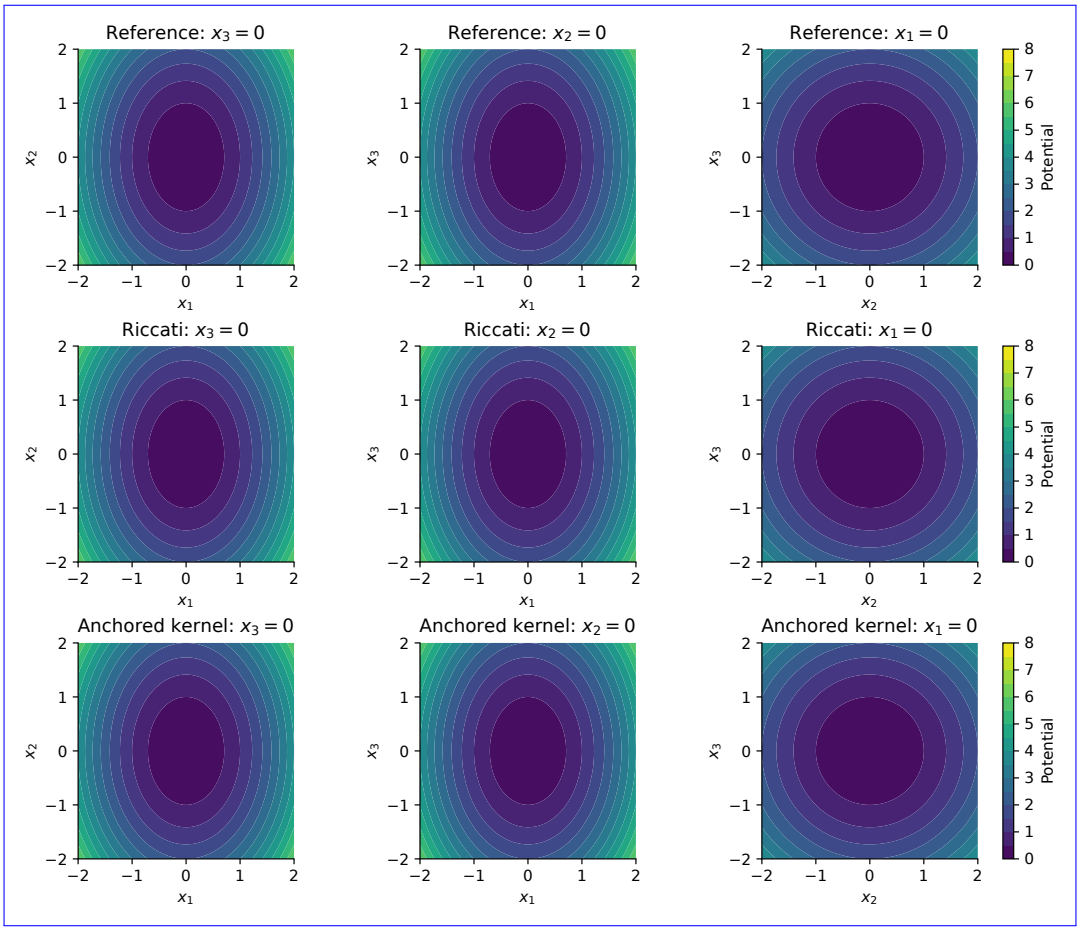


Figure 12: Coordinate-plane slices of the analytic potential, Riccati baseline, and anchored degree-two kernel fit at $N = 200$. Every fixed coordinate is exactly zero. A common potential range is used across all panels.

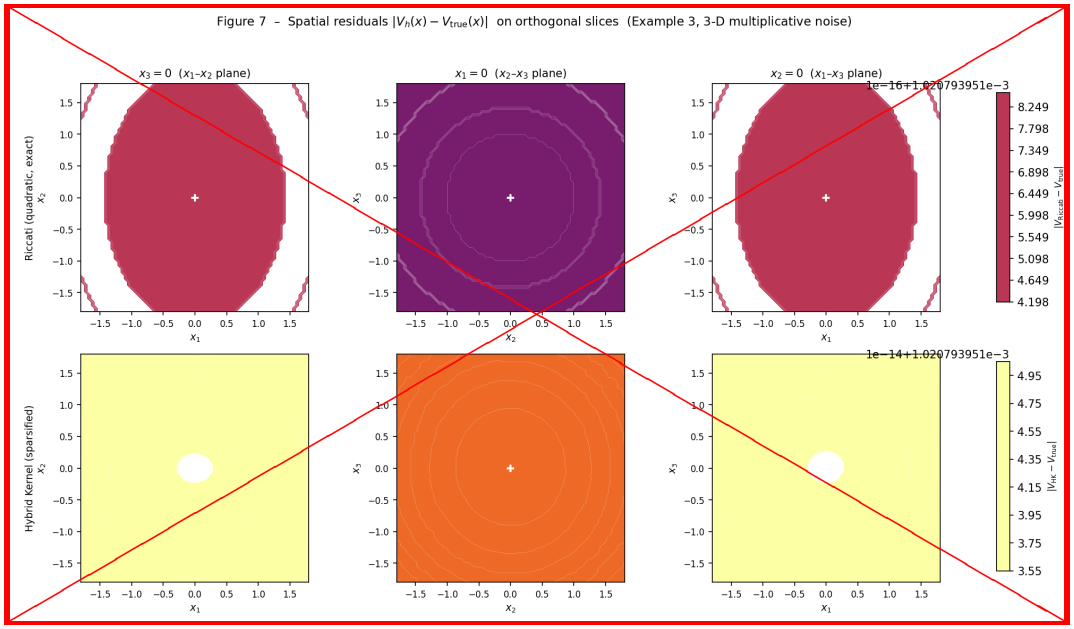


Figure 13: Spatial residuals $|V_h - V_{\text{true}}|$ for Example 3 on three orthogonal slices ($x_3 = 0$, $x_1 = 0$, $x_2 = 0$). Top row: Riccati (quadratic) approximation residuals, which achieve RMSE $\approx 10^{-3}$ since the true quasi-potential is exactly quadratic and the Riccati matrix captures this structure precisely. Bottom row: Hybrid Kernel approximation residuals, matching the Riccati accuracy (RMSE $\approx 10^{-3}$) after sparsification. Starting from a dictionary of 19 features (monomials up to degree 3), the ℓ_1 penalty automatically zeroes out all but three terms—precisely x_1^2, x_2^2, and x_3^2—recovering the true quadratic structure without prior knowledge of its form. Separate colorscales are used for each row; both are on the order of 10^{-3}, confirming that errors are negligible throughout the domain.

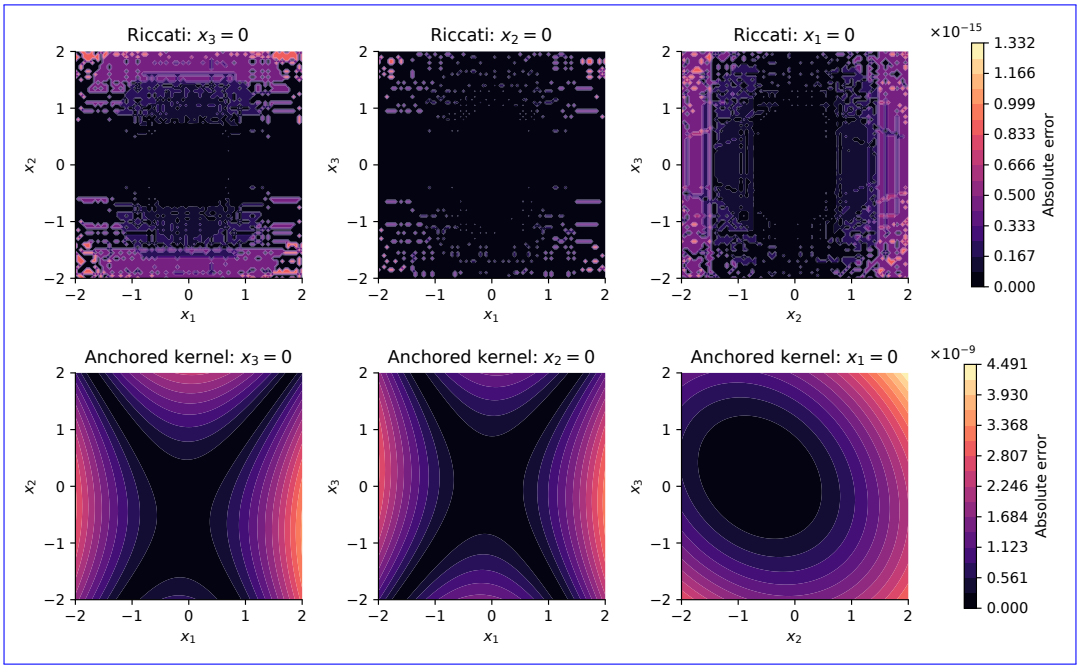


Figure 14: Absolute errors on the zero-coordinate slices in Figure 12. The top row is the Riccati baseline and the bottom row the anchored kernel fit. Each row uses its own labeled error scale.

5 Conclusion

We have presented a kernel-based framework for computing quasi-potentials in stochastic dynamical systems. The method parameterizes Koopman eigenfunctions via RKHS ansätze and minimizes a composite loss (18) that simultaneously enforces the kernel framework for quasi-potential computation with direct scalar and Koopman-coordinate representations. Both use Hamilton–Jacobi equation, matches the local Hessian to the Riccati solution, and promotes kernel sparsity through ℓ_1 regularization. The result is an analytical surrogate that is cheap to evaluate and automatically selects the correct functional form through sparsification. structure and local Riccati curvature. The spectral form supplies dynamical coordinates and a nonnegative quadratic surrogate, whereas direct scalar fitting avoids the additional transport restriction. Nonnegative amplitude regularization and discrete dictionary selection offer distinct controls on model complexity.

Theorem 3.2 establishes consistency of the learned surrogate with the true quasi-potential as the collocation count N grows, with convergence guaranteed in function values, gradients, and equilibrium curvature. Three benchmark examples validate the approach: Example 1 (gives sufficient conditions for convergence to the selected viscosity solution. The numerical experiments recover the two-dimensional gradient system) demonstrates kernel sparsification; Example 2 (double-well with non-trivial topology) shows that the hybrid surrogate improves upon the Riccati baseline; and Example 3 (three-dimensional multiplicative noise) confirms automatic recovery of the exact quadratic structure (RMSE $\approx 10^{-3}$, matching the Riccati baseline) and an empirical convergence rate RMSE $\sim N^{-0.73}$ consistent with kernel collocation theory quartic benchmark potentials using degree-four kernel spaces and the multiplicative-noise quadratic using an anchored degree-two expansion. Independent validation and test points assess residuals and potential errors. Finite-feature compression stabilizes the quadratic kernel refit without changing its function class. The controlled Gaussian-kernel experiments include a structured eight-dimensional system and favor direct fitting over the tested restricted spectral candidate. These findings identify when the global representational restriction, rather than the local curvature information, controls approximation accuracy.

Limitations and future work. The main limitation of the current work is the absence of a head-to-head comparison with established methods such as ordered-upwind methods [12, 20], string methods [24, 33], and geometric minimum action approaches [58]. Such a comparison, which would require careful tuning of each method on common test problems, is left as an important direction for future work. Computational cost scales as $O(N^2 n)$, limiting applicability beyond $n \approx 5$–8 dimensions with current implementations; Dense assembly and factorization have different costs, detailed in Section 4; the present evidence does not establish a universal practical dimension cutoff; sparse collocation, reduced-order modeling, and hierarchical kernel architectures are promising mitigation strategies. Extensions to non-Gaussian or Lévy noise, globally consistent multi-basin reconstruction, and the fully data-driven setting (where drift and diffusion must themselves be estimated) also remain open.

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A Controlled comparisons and cost measurements

A.1 Models and fitting protocol

This appendix compares direct and quadratic spectral representations and measures their computational costs. Both models use a single Gaussian kernel with bandwidth selection, coefficient ridge regularization, and exact local constraints. They complement the finite polynomial-kernel implementations in Section 4. The restricted quadratic Koopman candidate studied here is one specific spectral model; its performance does not characterize every hybrid kernel construction.

Let $y = x - \bar{x}$ and P be the Riccati Hessian. For kernel centers z_j, subtract Taylor jets at the equilibrium:

$$\psi_j^{(1)}(x) = k(x, z_j) - k(\bar{x}, z_j) - D_1 k(\bar{x}, z_j) y, \tag{33}$$

$$\psi_j^{(2)}(x) = \psi_j^{(1)}(x) - \tfrac{1}{2} y^\top D_1^2 k(\bar{x}, z_j) y. \tag{34}$$

These functions belong to an augmented kernel-plus-polynomial feature space. The direct scalar and restricted spectral candidates are

$$V_D(x) = \tfrac{1}{2} y^\top P y + \sum_{j=1}^{M} c_j \psi_j^{(2)}(x), \tag{35}$$

$$\Phi(x) = y + \sum_{j=1}^{M} c_j^K \psi_j^{(1)}(x), \qquad V_K(x) = \tfrac{1}{2} \Phi(x)^\top P \Phi(x). \tag{36}$$

Both satisfy $V(\bar{x}) = 0$, $\nabla V(\bar{x}) = 0$, and $D^2 V(\bar{x}) = P$ exactly. V_K is nonnegative; global nonnegativity of V_D is not automatic. The direct coefficients minimize mean squared HJ residual plus $10^{-8}\|c\|_2^2$. The spectral coefficients minimize the mean squared real vector transport residual $\|D\Phi\, b - F\Phi\|_2^2$ plus $10^{-8}\|C_K\|_F^2$. The latter is linear least squares in Mn coefficients. The Euclidean coefficient penalty differs from the RKHS norm defined by the kernel Gram matrix.

For each system and each seed in $\{230923, 230924, 230925\}$, the methods share $N = 256$ independent uniform training points, 512 validation points, 4096 test points, and the first $M = 32$ training points as centers. The kernel is Gaussian with $\ell/\sqrt{n} \in \{0.5, 1, 2\}$. Each method chooses its bandwidth by the smallest validation HJ residual; reference potential values never enter fitting or selection. Direct nonlinear least squares starts at the Riccati quadratic, uses an analytic Jacobian, tolerances 10^{-10}, and a 200-evaluation limit. The spectral fit uses the real coupled linear system. All candidate solves terminated successfully; the largest observed nonlinear evaluation count was 54. The comparison uses common data and constraints, but not equal parameter counts or identical fitting objectives.

A.2 Domains, higher-dimensional example, and results

The first three systems use the drift and diffusion coefficients in Section 4, with local testing boxes $[-1.5, -0.5] \times [-0.6, 0.6]$ for Example 1, $[0.5, 1.5] \times [-0.6, 0.6]$ for Example 2, and $[-1, 1]^3$ for Example 3. These local domains and Gaussian fitting protocols differ from the polynomial-kernel experiments in Section 4. Example 3's Riccati quadratic is exact and is the direct model's starting point; its direct result is therefore a consistency check, not a learned improvement over Riccati.

The eight-dimensional system is defined by

$$V_*(x) = \sum_{i=1}^{8}(x_i^2 + 0.1x_i^4), \quad b = -\tfrac{1}{2}\nabla V_* + R\nabla V_*, \quad a = I, \tag{37}$$

with $R_{i,i+1} = 1/4$, $R_{i+1,i} = -1/4$ and all other entries zero, on $[-1,1]^8$. Its equilibrium is zero, $F = -I+2R$ is Hurwitz, and $P = 2I$. Skew symmetry gives $H(x, \nabla V_*) = 0$. Completion of squares in the action gives $S_T \geq V_*(x) - V_*(0)$; the uphill flow $\dot{x} = \nabla V_*/2 + R\nabla V_*$ attains equality in the infinite-time limit. Its reverse decreases the coercive V_* strictly away from zero and converges to zero, which identifies the action value through truncation near the equilibrium. The drift is nonlinear and non-gradient, but the potential is separable, so this is a structured stress test rather than general high-dimensional validation.

Table 5: Test errors: mean ± sample standard deviation over three seeds. No reference-based rescaling is applied. MS, DW, Mult., and Rot. identify the four test systems.

Example	Model	RMSE	Relative ℓ^2 error	HJ RMS
1: MS, $d = 2$	Riccati	0.173 ± 0.00285	0.326 ± 0.00182	1.05 ± 0.0187
	Direct	0.00189 $\pm 3.10\times10^{-4}$	0.00355 $\pm 5.41\times10^{-4}$	0.00868 $\pm 8.01\times10^{-4}$
	Koopman	0.893 ± 0.0512	1.68 ± 0.119	33.6 ± 4.68
2: DW, $d = 2$	Riccati	0.0952 ± 0.00108	0.261 ± 0.00109	0.574 ± 0.00729
	Direct	0.0438 ± 0.0344	0.121 ± 0.0953	0.026 ± 0.00843
	Koopman	0.414 ± 0.0728	1.13 ± 0.186	13.1 ± 4.3
3: Mult., $d = 3$	Riccati	8.60×10^{-17} $\pm 1.22\times10^{-18}$	1.13×10^{-16} $\pm 8.75\times10^{-19}$	1.10×10^{-15} $\pm 3.10\times10^{-17}$
	Direct	8.60×10^{-17} $\pm 1.22\times10^{-18}$	1.13×10^{-16} $\pm 8.75\times10^{-19}$	1.10×10^{-15} $\pm 3.10\times10^{-17}$
	Koopman	0.195 ± 0.0207	0.256 ± 0.0286	0.351 ± 0.0145
4: Rot., $d = 8$	Riccati	0.177 $\pm 5.96\times10^{-4}$	0.0595 $\pm 1.07\times10^{-4}$	0.71 ± 0.00251
	Direct	0.0411 $\pm 1.86\times10^{-4}$	0.0138 $\pm 5.07\times10^{-5}$	0.147 ± 0.0051
	Koopman	0.485 ± 0.00386	0.163 ± 0.00162	1.46 ± 0.00471

Table 5 reports means and sample standard deviations across three seeds. Relative error is $\|u - V\|_{\ell^2(X_t)}/\|V\|_{\ell^2(X_t)}$; HJ RMS is the root mean squared residual on the independent test points. The direct model is more accurate than the restricted spectral candidate in all four tests. The double-well direct fit has appreciable seed variability, as shown by the tabulated standard deviation. The eight-dimensional mean relative errors are approximately 5.95% for Riccati, 1.38% for direct fitting, and 16.31% for the spectral candidate. These comparisons do not establish universal model rankings, a viscosity convergence certificate, or certified escape-barrier accuracy. They show why Koopman features should be treated as an optional restriction, rather than presumed to improve a direct fit.

A.3 Runtime, memory accounting, and reproducibility

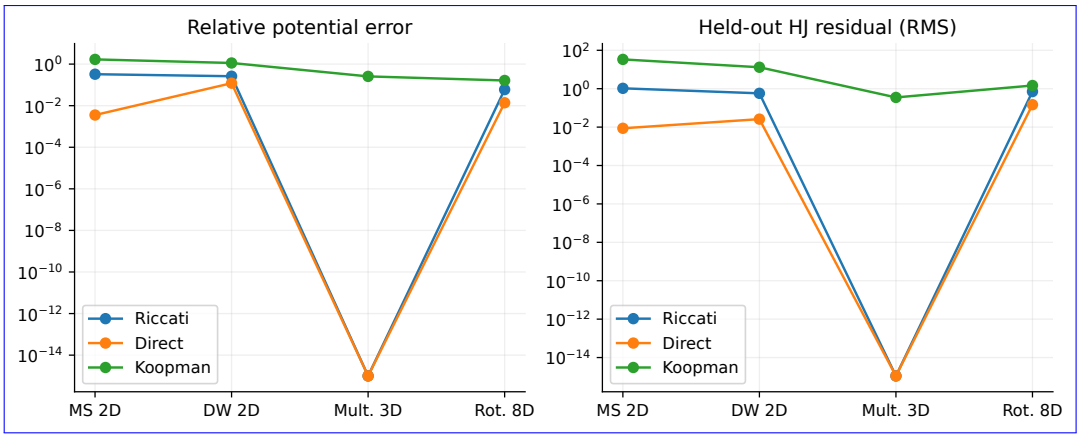


Figure 15: Controlled model comparison: mean relative potential error and independent-test HJ RMS over three seeds. MS, DW, Mult., and Rot. denote the Maier–Stein, double-well, multiplicative-noise, and eight-dimensional rotational systems. Values below 10^{-15} are drawn at 10^{-15} for visibility. All fits use the single-kernel protocol of Appendix A.1.

Table 6: Median measured time in milliseconds. Fit includes the three-bandwidth search and validation; test is value and gradient prediction on 4096 points. Problem setup, imports, plots, and error calculation are excluded.

	Direct		Koopman		Riccati
Example	Fit	Test	Fit	Test	Test
1: MS, $d = 2$	30.17	7.81	3.70	5.29	0.11
2: DW, $d = 2$	22.13	6.12	3.46	5.05	0.12
3: Mult., $d = 3$	4.87	7.70	5.09	6.56	0.16
4: Rot., $d = 8$	19.79	20.05	14.34	13.96	0.41

Table 7: Dense Gaussian Gram microbenchmark: median milliseconds after one warm-up, with storage for the Gram matrix alone. MiB denotes 2^{20} bytes. This is separate from the reduced-center fits.

d	N	Assembly (ms)	Cholesky (ms)	Gram (MiB)
2	128	0.105	0.037	0.125
2	256	0.277	0.271	0.500
2	512	1.731	1.748	2.000
2	1024	11.383	8.923	8.000
8	128	0.108	0.044	0.125
8	256	0.288	0.285	0.500
8	512	1.723	1.721	2.000
8	1024	12.268	10.436	8.000

Table 6 measures fitting/selection over all three bandwidths and value/gradient prediction at 4096 test points. Initial model/Lyapunov setup, imports, plots, and reference-error calculation are excluded. Table 7 separately measures Gaussian Gram assembly and Cholesky factorization, using $10^{-6}I$ regularization and medians of three repeats after one warm-up. It is not a timing of the complete PDE solver. Gram storage is $8N^2$ bytes; peak process memory was not measured and would also include factors, distances, derivatives, and optimization work arrays. A single 6400×6400 float64 matrix already occupies 312.5 MiB.

The recorded environment is CPU float64 on macOS 15.8 ARM64, Python 3.12.14, NumPy 2.5.3, SciPy 1.18.1, and Matplotlib 3.11.2. Single-thread limits were requested through environment variables and `threadpoolctl`; no inspectable pools were reported, so the vendor thread count was not independently confirmed. Short timings depend on system load and caching. Four matrix sizes do not establish a measured asymptotic complexity exponent.

The computational archive contains `code/run_experiments.py`, which defines the systems, derivatives, seeds, data splits, and solvers for the controlled comparisons. It checks gradients and equilibrium Hessians by finite differences and verifies the analytic HJ/Riccati identities and coupled transport assembly before fitting. The file `results/experiments.json` contains all 36 method/seed outcomes, candidate diagnostics, and timing/environment metadata, while `results/fitted_models.npz` stores coefficients, centers, and plotted grids. The scripts `code/make_tables.py` and `code/generate_figures.py` produce the tables and vector-PDF comparison plots from saved results. These materials support reproduction of the Gaussian-kernel comparisons. The main-text polynomial experiments are generated by `code/run_polynomial_experiments.py` and `code/refit_selected_kernel.py`; their saved inputs and outputs are in `results/accuracy/`. The program `code/plot_verified_experiments.py` generates the seven main-text figures and two numerical tables.

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